MAKING THE RIGHT CONNECTIONS: IMPROVING THE COMPREHENSION OF STRUGGLING READERS

GRANT# R324A110046
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I. ACCOMPLISHMENTS

A. Major Goals of the Project

The major goals of this project, as specified in the performance agreement, are to produce one or more fully developed questioning intervention(s) aimed at improving the reading comprehension of students at risk or identified as having disabilities. Intended products include: (a) one or more prototypes that are feasible and promising, with manuals; (b) one or more observational checklists to be used to document fidelity; and (c) training materials to prepare school-based personnel for implementation. We also aim to identify for whom (subgroups of struggling readers) and under what conditions (type and timing of questions, intensity of intervention) the intervention is most promising.

B. What was Accomplished under these Goals in Year 1

In Year 1, we completed or are in the process of completing the following activities to meet the above project goals: (1) Develop core intervention questions, materials, and procedures (*complete*); (2) Identify struggling readers to participate in pilot studies (*complete*); (3) Conduct pilot studies to examine feasibility and student outcomes (*complete*); (4) Analyze and interpret data from pilot studies (*preliminary analyses complete*); and (5) Refine interventions based on results of pilot studies (*in process*). These activities fall into three main categories: Identifying and Describing Struggling Readers, Text Selection and Development, and Intervention Development. Figure 1 provides details of activities under each category. Specific methods and results for each category are described in detail below.

Figure 1. ***Year 1 Timeline***

<table>
<tr><th></th><th colspan="2">Identifying and Describing Struggling Readers</th><th>Selecting Texts and Developing Questions</th><th>Developing Intervention</th></tr>
<tr><td>Sep-11</td><td colspan="2">Screening Phase 1</td><td rowspan="11">Cycle of Identifying Potential Texts, Transcribing, Determining Text Ease/ Readability, Developing & Reviewing Questions and Possible Responses</td><td rowspan="2">Round 1: In-House Development</td></tr>
<tr><td>Oct-11</td><td colspan="2" rowspan="2">Screening Phase 2 & Think-Aloud Administration</td></tr>
<tr><td>Nov-11</td><td>Round 2: Focus Groups</td></tr>
<tr><td>Dec-11</td><td rowspan="6">Think-Aloud Coding</td><td></td><td>Round 3: Fall Trial Run</td></tr>
<tr><td>Jan-12</td><td rowspan="2">Descriptive Testing</td><td rowspan="3">Round 4: Spring Pilot</td></tr>
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<tr><td>Mar-12</td><td></td></tr>
<tr><td>Apr-12</td><td></td><td>(scoring and data entry)</td></tr>
<tr><td>May-12</td><td></td><td>Round 5: Spring Trial Run</td></tr>
<tr><td>Jun-12</td><td colspan="2">Identification of Subgroups</td><td></td><td>(Spring Pilot recall coding)</td></tr>
<tr><td>Jul-12</td><td></td><td></td><td></td><td>(data analysis)</td></tr>
<tr><td>Aug-12</td><td></td><td></td><td>Revision of Texts and Questions</td><td>Round 1b: In-House Revision of Intervention</td></tr>
</table>

Identifying and Describing Struggling Readers

Purpose and Research Questions

The purpose of identifying and describing struggling readers was to (a) identify students struggling primarily with comprehension, and thus eligible for the intervention to be developed, (b) identify subgroups of struggling comprehenders, and (c) describe characteristics of struggling readers (and, eventually, identify potential moderators to students' response to the intervention). Specific research questions include:

(1) Of a diverse sample of fourth-grade students, which students are struggling primarily with reading comprehension (not decoding/word recognition or fluency skills)?
(2) Of those students identified as struggling comprehenders, do subgroups of comprehenders emerge, based on responses to a think-aloud task?
(3) If subgroups of comprehenders emerge, do the subgroups differ on reading-related variables other than the think-aloud task?

Method

Participants

Districts and schools. Approval to conduct this research was obtained from two districts' Departments of Research, Evaluation & Assessment (REA) prior to recruiting teachers and students. District 1 is a large urban district serving about 32,263 K-12 students in 70 schools at the time of this study (2011-12 school year). District 2 is a first-ring suburb of a large urban area serving 10,313 K-12 students in 15 schools at the time of this study. A total of seven schools across the two districts agreed to participate. District and school demographic information for the study year (2011-12) is summarized in Table 1.

Table 1. Participating District and School Demographic Information

	Total Enrollment	Students of Color (%)	English Learners (%)	Free/Reduced Lunch (%)	Special Ed (%)
District 1	32,263	68	21	66	19
School 1	588	94	65	97	14
School 2	650	24	1	17	12
School 3	376	43	1	37	13
School 4	668	16	1	11	9
District 2	10,313	37	4	33	12
School 5	525	42	18	34	8
School 6	581	41	18	34	13
School 7	544	86	44	80	9

The two districts preferred different methods for recruiting schools and teachers. In District 1, we contacted principals and teachers directly. The PI first contacted 4th-grade teachers in the district via an email that included a brief overview of the study. For those teachers who responded to the email indicating interest, the PI arranged a 30-min initial meeting with the teachers. In all cases, the PI met with whole 4th-grade teams from a building. In some cases,

reading specialists and/or special education teachers also attended the meetings. In the initial meeting, the PI provided an information sheet and reviewed the study purpose and procedures. Teachers either expressed interest on the spot or within approximately one week. School principals were then contacted to confirm their approval of the teachers' participation.
In District 2, the PI first met with district REA and curriculum directors to explain the study, then met with the district's literacy leadership team and described the study using the information sheet. School representatives returned to their buildings and presented the information to principals and 4th-grade teams. Principals then submitted names of interested teachers to the curriculum director, who forwarded them to the PI.

Teachers. Thirty-two teachers participated, including 23 fourth-grade classroom teachers and 9 specialists. Specialists described their positions and job responsibilities as follows: 2 reading resource teachers (taught reading/implemented guided reading and provided reading intervention), 4 literacy coaches (provided professional development to teachers, administered reading assessments, supported reader's/writer's workshop model), 2 special education teachers (1 multi-categorical teacher, 1 D/HH teacher; maintained caseloads, provided intervention and RTI support), 1 EL teacher (served ELs in grades 3/4, provided reading & writing support in collaborative and pull-out models). Teacher-reported demographic data are in Table 2.

Table 2. ***Teacher-Reported Demographics***

	District 1 (*n* = 17)				District 2 (*n* = 15)			
	Classroom Teachers (*n* = 11)		Specialists (*n* = 6)		Classroom Teachers (*n* = 12)		Specialists (*n* = 3)	
	n	%	*n*	%	*n*	%	*n*	%
Sex (female)	8	72.7	6	100.0	10	83.3	3	100.0
Race								
White	10	90.9	6	100.0	10	83.3	2	66.7
Hispanic	1	9.1	0	0.0	0	0.0	1	33.3
Other	0	0.0	0	0.0	2	16.7	0	0.0
Age								
20-29	0	0.0	0	0.0	3	25.0	0	0.0
30-39	2	18.2	0	0.0	6	50.0	2	66.7
40-49	5	45.5	3	50.0	2	16.7	1	33.3
50-59	3	27.3	1	16.7	1	8.3	0	0.0
60+	1	9.1	2	33.3	0	0.0	0	0.0
Highest Degree								
Bachelor's	1	9.1	0	0.0	3	25.0	0	0.0
Master's	3	27.3	0	0.0	5	41.7	1	33.3
Master's +	7	63.6	6	100.0	4	33.3	2	66.7
	M (range)	*SD*	*M* (range)	*SD*	*M* (range)	*SD*	*M* (range)	*SD*
Years in current position	8 (1-28)	8.8	6 (1-18)	6.1	5.55 (1-16)	5.2	10 (7-12)	2.7
Years in general education	17.64 (7-29)	6.9	21.17 (5-38)	11.8	11.27 (5-25)	6.0	6 (0-18)	10.4
Years teaching Grade 4	9.64 (0-33)	9.9	2.33 (0-12)	4.8	4.5 (0-13)	4.3	1.67 (0-5)	2.9
Years teaching special education	1 (0-11)	3.3	2.83 (0-12)	4.9	.08 (0-1)	0.3	10.67 (0-20)	10.1
Years as a reading specialist	.27 (0-2)	0.7	9 (0-25)	10.8	.17 (0-2)	0.6	1.33 (0-4)	2.3
Students in class/caseload	32.09 (30-34)	1.5	75.5 (0-250)	118.6	24.58 (19-30)	3.6	34.67 (23-55)	17.7

Students. All fourth-grade students in participating classrooms were invited to participate in the study. Research staff provided brief (5-10 min) presentations of the study to each participating class, and distributed consent forms to all students to take home and have signed by their parents. Students were given gel pens when they returned the signed consent forms (whether the parents indicated 'yes' or 'no' to participation). Table 3 shows the distribution and return rates of parental consent forms in the two districts.

Table 3. ***Distribution and Return Rates of Consent Forms***

	Distributed	Returned Indicating 'Yes' or 'No' to Participation			Return Rate
		Yes	No	Total	
District 1	336	251	45	296	88%
District 2	274	218	33	251	92%
Total	610	469	78	547	90%

Measures

Measures for identifying and describing participants fall into the following three categories: screening measures, think-aloud task, and descriptive measures.

Screening measures. Screening occurred in two phases. Phase I screening measures included CBM Maze and district tests. Phase II screening measures included CBM oral reading. Table 4 shows intercorrelations among screening measures.

Curriculum-Based Measurement (CBM)—Maze Task. The CBM maze task (e.g., Fuchs & Fuchs, 1992) is designed to be a brief assessment of overall reading proficiency. The maze is group administered, and takes about 10-15 min to administer. In this study, students read two passages in which every seventh word was deleted and replaced with three word choices, only one of which fit in the sentence. Students completed two practice sentences, and then read each passage silently for 2.5 min, selecting the words that best fit in the sentences. The score was the average correct maze selections made on the two passages. Alternate-form reliability coefficients have been reported to range from $r = .83$ to .98, and correlations between maze selection scores and other reading measures typically range from $r = .60$ to .80 for elementary-aged students (Wayman, Wallace, Wiley, Ticha, & Espin, 2007). In the screening sample for this study ($n = 465$), test-retest reliability for the maze was $r = .88$, $p < .001$.

District tests. District 1 used the Computerized Achievement Levels Test (CALT) and District 2 used the Measures of Academic Progress (MAP). Both are developed by the Northwest Evaluation Association and assess students' reading achievement and growth from late elementary through high school. Reading achievement is assessed in Minnesota in the areas of "Literature," "Informational Text," "Foundational Skills and Vocabulary" and "Language Use and Structure." These areas are aligned with district and state reading standards.

All MAP assessments are computerized and are presented adaptively, adjusting in difficulty based on each student's responses to test questions. They commonly consist of more

than 40 questions. Assessments are provided in a survey structure with content area goal scores supplementing an overall score. For the purposes of this study, we recorded the Normal Curve Equivalent (NCE) and RIT score, which allows the values of achievement levels and the values of items to be on the same scale (Northwest Evaluation Association, 2008).

CBM oral reading. The CBM oral reading task (Deno, 1985) consists of individually-administered grade-level passages, which students read aloud for 1 min. As the student reads, the examiner marks insertions, omissions, substitutions, and mispronunciations not caused by speech-related problems or dialects. Errors are not corrected and no decoding assistance is given. In this study, the student read two passages; the score is the mean number of words read correctly in one min. For this study, we used two Formative Assessment System for Teachers (FAST; Christ, Ardoin, Eckert, & White, 2010a) 4th-grade-level CBM passages. Criterion validity for the 4th-grade passages with other reading measures (DIBELS and TOSREC) has been reported as r = .95 (DIBELS) and .74 (TOSREC), and classification accuracy for identifying students at risk has been reported as strong (Christ, Ardoin, Eckert, & White, 2010b). In the screening sample for this study (n = 132), test-retest reliability was r = .92, p < .001.

Table 4. ***Intercorrelations Among Screening Measures***

	1	2	3
1. Maze			
2. CALT	.71 (n = 217)		
3. MAP	.77 (n = 222)	na	
4. CBM Oral Reading	.71 (n = 134)	.78 (n = 65)	.75 (n = 66)

Note. CALT = Computerized Achievement Levels Test, MAP = Measures of Academic Progress. *N*s vary due to (a) CALT/MAP given in different districts and (b) CBM Oral Reading given only to students in Screening Phase II.

Think-aloud task. The think-aloud task was used to identify possible subgroups of struggling comprehenders. As in previous related studies (e.g., McMaster et al., 2012; Rapp et al., 2007), we used a think-aloud task adapted from Ericsson & Simon (1993), which included a practice text ("Jimmy and the New Bike," adapted from Magliano, Trabasso, & Graesser, 1999), two narrative texts ("Brian's Magical Skill" and "Candle Crafts for School" and two informational texts ("Living Underground: The Life of a Mole" and "Mountain Heights" adapted from van den Broek et al., 2001). Each sentence from each text was typed on 5" X 6" cards. Texts are provided in Appendix A.

The examiner demonstrated the think-aloud task with the first half of the practice text by giving specific examples of all possible types of responses (e.g., elaborative inferences, paraphrases, predictions). Then, the student practiced thinking aloud with the remaining sentences in the practice text. The student read one sentence at a time and then told the examiner what he or she was thinking. After each text, the examiner asked the student two yes/no questions about the story to check for general comprehension. The student received no decoding help, but received non-leading prompts such as, "What are you thinking now?" if he or she forgot to think aloud.

Descriptive measures. In January, 2012, before intervention began, we administered a battery of descriptive tests to all students identified as eligible for the intervention. This battery

included the Letter-Word Identification, Word Attack, Reading Vocabulary, and Academic Knowledge subtests of the Woodcock-Johnson Tests of Achievement-III (WJ III ACH; Woodcock, Mather, & McGrew, 2001), and the Motivation for Reading Questionnaire (MRQ; Wigfield & Guthrie, 1997) to more completely describe the struggling readers, as well as to determine if specific factors moderate students' comprehension of different text genres under varying intervention conditions.

WJ III ACH subtests. The WJ III ACH (Woodcock et al., 2001) Letter-Word Identification subtest requires the student to name letters and read words aloud from a list, the Word Attack subtest involves reading nonsense words aloud to test phonetic word attack skills, the Reading Vocabulary subtest involves asking the student to orally state synonyms and antonyms for printed words and orally complete written analogies, and the Academic Knowledge subtest involves asking the student oral questions about factual knowledge of science, social studies, and humanities. Median reliability coefficients have ranged from .81 to .94, and validity coefficients have ranged from 0.50 to 0.80 with corresponding subtests of the Kaufman Test of Educational Achievement (Kaufman & Kaufman, 2004) and the Wechsler Individual Achievement Test (Wechsler, 2001).

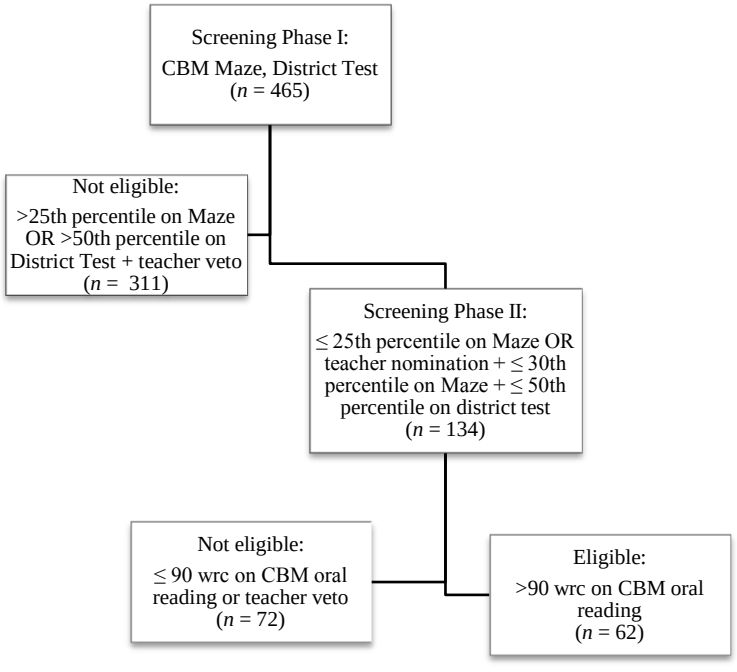
Motivation for Reading Questionnaire (MRQ). The MRQ (Wigfield & Guthrie, 1997) assesses 11 aspects of reading motivation: reading efficacy, reading curiosity, reading involvement, importance of reading, reading challenge, reading work avoidance, competition in reading, recognition of reading, reading for grades, social reasons for reading, and compliance. Only the first four aspects were assessed as part of this study, on the recommendation of Guthrie (personal communication, January 2012).

The MRQ is an independent, group-administered, self-report measure of reading motivation. The full 11 construct MRQ contains 53 items and takes 15 to 20 minutes to administer. The 4-construct version used in this study included 17 items and took approximately 5 min to administer. As a group, students answered two practice questions which asked them to identify to their level of agreement with a statement (e.g. "I like ice cream") using a four point rating scale (1 = very different than me to 4 = A lot like me). Students then completed the 17 MRQ items independently, while the teacher read each statement out loud to the group.

Procedures

Screening. To determine eligibility for the intervention, all consented students (n = 465) were screened in two phases. In Screening Phase I, conducted in September, 2011, research staff administered two CBM Maze probes and collected district test data (CALT or MAP). Students whose average scores from the two CBM Maze probes were at or below the 25th percentile were identified as *potentially eligible* for Screening Phase II. The bottom 25th percentile of both districts was 14 correct maze choices (cmc) (District 1 = 14, District 2 = 14.5); we used the full sample percentile. Students were identified as not eligible or eligible for Screening Phase II using a combination of Maze, district test data, and teacher judgment, as illustrated in Figure 2.

Figure 2. ***Participant Flow through Screening Process***



Those students identified as eligible for Screening Phase II, conducted in October-November, 2011, were individually administered two CBM oral reading probes. Those whose mean score was less than or equal to 90 words read correctly (wrc) in one min, or whose teachers felt they would not benefit from intervention due to poor decoding or English language skills, were identified as ineligible for intervention (n = 47; 10% of sample), because their reading difficulties were likely at least partly due to decoding or word recognition problems (Christ et al., 2010b). Those who scored greater than 90 wrc in one min were identified as eligible for the intervention (n = 62; 13%). (One child scored lower than 90 wrc but was included due to teacher nomination.)

Think-aloud administration and coding. Students identified as eligible for the intervention based on CBM oral reading scores were also administered think-aloud tasks to identify subgroups of struggling readers. The think-aloud task was conducted using a script within the "CBM Oral Reading and Think Aloud" directions. The Discourse Project Coordinator trained GRAs to administer the think-aloud task by:

1. Modeling the procedure.
2. Observing each GRA walk through the procedure accordint to the script, using the story cards, and audiorecording the session.
3. Verifying proficiency of administration by the GRA after a period of independent practice.

Think-aloud tasks were administered individually during Screening Phase II, using two narrative and two informational texts. GRAs were instructed to only administer the think-aloud task if the student had scored at least 65 words read correctly on at least one of the CBM oral reading probes. The entire session was audio recorded and transcribed.

Undergraduate Research Assistants (URAs) transcribed verbatim the think-aloud audiorecordings according to a written procedure. Twenty percent of the transcripts, including at least one for each URA transcriber, were verified for accuracy by the Discourse Project Coordinator or GRA. Think-aloud transcripts were then parsed into thought units, which represent a complete idea with subject and verb, and possibly subordinate clause**.** Each parsed thought unit was categorized based on a coding scheme adapted from previous research (Linderholm & van den Broek, 2002) designed to identify the cognitive processes in which readers engaged during reading. The primary variables coded included:

- *Elaborative inferences* (readers' attempts to explain the contents of the current sentence using background knowledge. Coded as *valid* when explanations were constrained by the text, plausible and accurate, or *invalid* when explanations were not constrained by the text, were implausible or reflected a misunderstanding or other error)
- *Predictive inferences* (readers' anticipations of what would occur next in the text, also coded as *valid* or *invalid* based on whether the predictions were constrained by the text, were plausible or involved error)
- *Connecting inferences* (readers' explanations of the current sentence by connecting it to the immediately preceding sentence)
- *Reinstatement inferences* (readers' explanations of the current sentence on the basis of prior text information not in the immediately preceding sentence)
- *Associations* (readers'comments about the current sentence that draw upon background knowledge as an aside
- *Paraphrases* (readers' rephrasing of the current sentence in their own words)
- *Text repetitions* (readers' verbatim reiterations of the current sentence)
- *Metacognitive comments* (readers' reflections on their understanding of text. Coded as *agreement, disagreement,* or *no prior knowledge* based on the substance of the comment)
- *Evaluative comments* (readers' opinions about the text. Coded as evaluating the *function* or *content* of the text)
- *Affective responses* (readers' reflections of their own emotions in response to the current sentence)
- *Questions* (readers' questions asked about the content of the current sentence).
- *Other* (readers' statements that can not be categorized often because of inaudible utterances)

After coding all transcripts with these finely delineated codes, the codes with low frequencies were collapsed with related code(s) into a new category. All metacognitive, evaluative and affective comments (MCA, MCB, MCC, Ea, Eb, AR) were collapsed into a single "metacognitive comment" code. Likewise, because the text repetition code frequency was so low, text repetitions were collapsed with paraphrases into a single "text paraphrase" code. "Other" codes were excluded from analysis. Kappa calculations were based on the collapsed codes. Original codes, categories, frequencies, and final collapsed codes are in Table 5.

Table 5. ***Frequencies of Think-Aloud Codes, and Fnal Collapsed Codes***

ORIGINAL CODE	NAME OF CATEGORY	FREQUENCY	FINAL CODE
EII	Elaborative Inference Invalid	574	EII
EIV	Elaborative Inference Valid	1706	EIV
PII	Predictive Inference Invalid	48	PII
PIV	Predictive Inference Valid	470	PIV
CI	Connecting Inference	209	CI
RI	Reinstatement Inference	443	RI
A	Association	381	A
P	Paraphrase	1218	TP
TR	Text Repetition	45	
MCA	Metacognitive Comment - Agreement	131	MC
MCB	Metacognitive Comment - Disagreement	53	
MCC	Metacognitive Comment - No prior knowledge	24	
Ea	Evaluative - function	1	
Eb	Evaluative -content	26	
AR	Affective Response	4	
Q	Question	222	Q
O	Other	39	O (not analyzed)

The think-aloud transcripts were coded by trained URAs. The Discourse Project Coordinator and advanced GRA practiced coding until their IRA approached 80% (ranged from 76% to 81%). Coding training, practice and IRA proceeded as follows:

1. *Two-part training session*. An experienced think-aloud expert (Dr. Sarah Carlson) taught the purpose of think-aloud coding, the definitions of each code, rules for coding, and provided guided practice.
2. *Group coding practice*. Project coordinator, advanced GRA and two or more URAs practiced applying the think-aloud coding rules and procedures to actual think-aloud transcripts.
3. *Paired coding practice*. Pairs of URAs coded transcripts together until they were comfortable with their level of agreement.
4. *Individual coding with every third transcript subjected to consensus coding*. Pairs coded transcripts in sets of five; three each were coded individually, and one transcript was reviewed for consensus and IRA calculated.
5. *Third party IRA was conducted for every pair of coders*. If IRA was less than 75%, all subsequent transcripts were recoded.

Thirty-four percent of all transcripts were coded in practice under the direction of the project coordinator or advanced GRA. Third party IRA was conducted by the project coordinator or advanced GRA on 18% of the 38 transcripts coded by individual URAs or coded in the consensus process by pairs of URAs. URAs whose IRA with the project coordinator or advanced GRA was low (below 75%) had their transcripts re-coded.

Kappa was calculated for agreement between coding pairs, and between the third party and coding pairs and individuals. Kappa ranged from .45 to .85. URAs with Kappa under 50% had their transcripts recoded by a project coordinator or advanced GRA.

Descriptive test administration and scoring. The WJ III ACH subtests were administered in mid-January, 2012. Six GRAs and three URAs were trained to administer the measures in one 2-hour session. During training, examiners learned the purpose of administering the subtests, the importance of standardized administration, how to use basal and ceiling rules, and how to administer the subtests including using the materials and prompting procedures. Following training, examiners practiced administering the measures, and then met individually with the PI or one of two GRAs on the project who were experienced in administering the WJ III ACH (one was a former special education teacher and one was a school psychologist, both of whom had administered the measures to numerous students). The purpose of the meeting was to 'check out'--the examiner administered all subtests to the experienced tester, who gave canned student answers and completed the checklist provided in the testing manual of necessary test administration components. Feedback was given regarding the examiner's administration. The examiner then scored the testing protocol, and his or her scoring was compared with the correct scoring. If the examiner was not in at least 90% agreement with the correct scoring on a given subtest, he or she was required to redo the check out for that subtest, until reaching criterion. Once criterion was reached, the examiner was deemed ready to administer the subtests.

During testing, each examiner spent several minutes establishing rapport with the student, and then administered the four subtests in random order to avoid an order effect.

The MRQ was administered in late January, 2012 as part of the introductory session to the intervention. Examiners were trained to provide the standardized instructions, and asked to monitor students closely to make sure they answered all of the questions. The MRQ was administered in the students' small intervention groups (3-5 students per group).

Results

Research Question 1: Of a diverse sample of fourth-grade students, which students are struggling primarily with reading comprehension (not decoding/word recognition or fluency skills)?

This question was addressed through the screening process described above. Table 6 shows descriptive data obtained from Screening Phases I and II as well as for students identified as eligible for intervention.

Table 6. ***Descriptive Statistics from Screening Phases I and II***

	Screening Phase I: Full Sample (n = 465)				
	n	Mean	*SD*	Min	Max
District 1					
Maze	248	19.32	7.57	2.50	38.50
District SS	222	204.68	16.84	153.00	237.00
District Percentile	223	61.56	31.56	1.00	99.00
District 2					
Maze	217	19.15	7.60	0.50	43.00
District SS	217	203.01	13.98	152.00	234.00
District Percentile	217	59.12	28.78	1.00	99.00

	Eligible for Screening Phase II (n = 134)				
	n	Mean	*SD*	Min	Max
District 1					
Maze	69	10.37	3.59	2.50	15.50
District Test SS	66	186.02	14.61	153.00	216.00
District Test Percentile	66	25.38	25.07	1.00	87.00
Oral Reading	69	101.20	30.24	13.00	164.00
District 2					
Maze	65	10.99	3.92	0.50	15.50
District Test SS	65	191.62	15.04	152.00	223.00
District Test Percentile	65	35.15	26.38	1.00	97.00
Oral Reading	65	103.58	31.63	34.00	166.00
	Eligible for Intervention (n = 62)				
	n	Mean	*SD*	Min	Max
District 1					
Maze	30	10.97	3.05	4.00	15.00
District Test SS	29	190.07	10.21	174.00	209.00
District Test Percentile	29	28.97	22.11	4.00	74.00
Oral Reading	30	114.75	19.04	83.00	158.00
District 2					
Maze	32	11.66	2.58	4.50	15.50
District Test SS	32	193.75	11.52	154.00	208.00
District Test Percentile	32	36.94	21.22	1.00	72.00
Oral Reading	32	114.33	16.96	92.00	153.00

Research Question 2: Of those students identified as struggling comprehenders, do subgroups of comprehenders emerge, based on responses to a think-aloud task?

To identify possible subgroups of struggling readers, think-aloud data were submitted to a cluster analysis using the *k*-means cluster approach for both the narrative and informational texts. We attempted both two- and three-group solutions to determine the best fit for the current data. In previous studies, a two-group solution best fit the data. However, in this study, for both the narrative and informational texts, a three-group solution yielded potentially meaningful information. Specifically, when a three-group solution was used, two expected groups emerged: one that was primarily distinguished by paraphrasing (i.e., 'paraphrasers,' n = 16 and 18 for narrative and informational) and one that was primarily distinguished by elaborating (i.e., 'elaborators,' n = 44 and 42 for narrative and informational). The two groups emerged in the expected proportions based on previous research (1.25:1 to 1.90:1 with more Elaborators than Paraphrasers; e.g., McMaster et al., 2012; Rapp et al., 2007; van den Broek et al., 2006). Further, a majority of students (94%) were identified as being in the same subgroup for both the narrative and informational think-aloud tasks. This finding is noteworthy given that a goal of this project was to determine whether the think-aloud approach yielded consistent subgroups using informational text as it did using narrative text.

For both the narrative and informational text, a third group emerged that we have not observed in previous research. This group had only two students, whose think-alouds were distinguished by use of questioning as their primary type of think-aloud statement. The same two students emerged in both the narrative and informational think-aloud clusters. We re-examined these two students' think-aloud protocols to make sure they were coded correctly, and determined that they were. We also re-ran the cluster analyses without these two students, considering that they were outliers, but eliminating them did not change the two-group solution,

and the three-group solution was not conceptually meaningful. Further, we determined that it is important to examine these two students' descriptive data and responsiveness to intervention to determine whether their unique subgroup status might be meaningful. Thus, we accepted the three-group solution using all participating students. Table 7 provides descriptive data illustrating the three groups' proportions of each type of think-aloud category in both narrative and informational text.

It is important to note that we were not able to identify subgroups of struggling readers before intervention began due to the time needed to code the data. Coding took longer than expected, in part because the original project coordinator with expertise in this area had to leave the project, and a steeper-than-expected learning curve was required to conduct the coding reliably and accurately. In future years of the project, the aim will be to identify subgroups prior to assignment to intervention condition.

Table 7. ***Proportions of Think-Aloud Statements made by Three Subgroups***

	Narrative Think-Aloud					Informational Think-Aloud				
	Mean	*SD*	*F* (2, 59)	*p*		Mean	*SD*	*F* (2, 59)	*p*	
Associations			3.50	.037	P > E			3.12	.051	P > E
Paraphrasers	0.005	0.010				0.034	0.057			
Questioners	0.000	0.000				0.058	0.082			
Elaborators	0.042	0.059				0.106	0.117			
Total	0.031	0.053				0.083	0.107			
Connecting Inferences			1.76	.181	ns			1.67	.197	ns
Paraphrasers	0.049	0.036				0.012	0.018			
Questioners	0.013	0.018				0.000	0.000			
Elaborators	0.056	0.033				0.023	0.028			
Total	0.053	0.034				0.019	0.025			
Reinstatements			6.75	.002	ns			2.28	.111	ns
Paraphrasers	0.065	0.065				0.014	0.024			
Questioners	0.013	0.018				0.000	0.000			
Elaborators	0.149	0.098				0.031	0.035			
Total	0.123	0.098				0.025	0.033			
Elaborative Inferences - Invalid			0.81	.449	ns			1.09	.341	ns
Paraphrasers	0.057	0.067				0.145	0.112			
Questioners	0.000	0.000				0.028	0.007			
Elaborators	0.061	0.068				0.167	0.145			
Total	0.058	0.067				0.156	0.135			
Elaborative Inferences - Valid			25.75	.000	E > P			22.56	.000	E > P
Paraphrasers	0.113	0.075				0.179	0.088			
Questioners	0.059	0.022				0.035	0.049			
Elaborators	0.361	0.142				0.411	0.156			
Total	0.288	0.171				0.332	0.181			
Predictive Inferences - Invalid			1.38	.259	ns			1.24	.296	ns
Paraphrasers	0.006	0.013				0.001	0.006			
Questioners	0.000	0.000				0.000	0.000			
Elaborators	0.013	0.021				0.006	0.013			
Total	0.011	0.019				0.005	0.012			
Predictive Inferences - Valid			5.25	.008	ns			1.60	.211	ns
Paraphrasers	0.031	0.033				0.038	0.022			
Questioners	0.011	0.015				0.000	0.000			
Elaborators	0.143	0.147				0.055	0.059			
Total	0.110	0.135				0.048	0.051			
Metacognitive Comments			1.11	.337	ns			1.37	.262	ns
Paraphrasers	0.008	0.014				0.042	0.048			
Questioners	0.022	0.031				0.000	0.000			
Elaborators	0.020	0.031				0.075	0.102			
Total	0.017	0.027				0.063	0.089			
Text Repetitions/Paraphrases			92.49	.000	P > Q, E			106.06	.000	P > Q, E
Paraphrasers	0.658	0.168				0.504	0.132			
Questioners	0.000	0.000				0.023	0.033			
Elaborators	0.143	0.120				0.109	0.081			
Total	0.272	0.265				0.221	0.207			
Questions			1521.04	.000	Q > P, E			288.12	.000	Q > P, E
Paraphrasers	0.008	0.020				0.030	0.069			
Questioners	0.883	0.012				0.855	0.157			
Elaborators	0.009	0.023				0.017	0.028			
Total	0.037	0.157				0.048	0.156			

Note. For narrative, Paraphrasers *n* = 16, Questioners *n* =2, Elaborators *n* =44. For informational , Paraphrasers *n* =18, Questioners *n* =2, Elaborators *n* =42.

Research Question 3: If subgroups of comprehenders emerge, do the subgroups differ on reading-related variables other than the think-aloud task?

Table 8 provides descriptive statistics of students' performance on these measures by subgroup (as well as by Causal and General intervention group, described later in this report). There were no statistically significant between-group differences on any of these measures.

Table 8. ***Subgroups' Performance on Descriptive Measures***

	Causal			General		
	Paraphrasers (*n* = 7)	Elaborators (*n* = 22)	Total (*n* = 29)	Paraphrasers (*n* = 9)	Elaborators (*n* = 21)	Total (*n* = 30)
WJ-III ACH						
Word ID Raw						
Mean	49.29	48.50	48.69	49.11	49.05	49.07
SD	5.35	3.84	4.16	5.88	5.89	5.79
Word ID SS						
Mean	98.43	97.50	97.72	98.44	98.19	98.27
SD	7.93	5.71	6.17	7.97	9.64	9.03
Word Attack Raw						
Mean	19.00	20.23	19.93	18.78	18.95	18.90
SD	4.76	4.03	4.17	5.95	5.51	5.54
Word Attack SS						
Mean	97.43	99.00	98.62	97.33	97.48	97.43
SD	6.90	6.64	6.61	9.03	8.21	8.31
Reading Vocab Raw						
Mean	27.14	23.82	24.62	23.56	24.24	24.03
SD	3.63	3.90	4.04	6.71	6.62	6.54
Reading Vocab SS						
Mean	96.43	90.91	92.24	91.44	91.71	91.63
SD	6.40	4.98	5.76	9.72	11.25	10.65
Academic Knowledge Raw						
Mean	43.00	40.27	40.93	37.67	41.43	40.30
SD	1.91	4.08	3.84	7.83	4.51	5.83
Academic Knowledge SS						
Mean	89.71	81.77	83.69	74.67	85.81	82.47
SD	7.27	9.51	9.55	22.11	14.34	17.43
Motivation for Reading Questionnaire						
Reading Involvement						
Mean	17.29	17.57	17.50	19.89	19.43	19.57
SD	4.39	3.89	3.94	4.11	4.82	4.55
Reading Efficacy						
Mean	9.57	10.26	10.10	9.00	10.05	9.73
SD	1.81	1.86	1.84	2.87	2.31	2.49
Reading Curiosity						
Mean	16.57	18.96	18.40	19.44	17.95	18.40
SD	3.46	3.51	3.59	3.43	3.76	3.67
Importance of Reading						
Mean	6.43	6.87	6.77	7.22	7.33	7.30
SD	1.40	1.60	1.55	0.97	1.20	1.12
Reading Total						
Mean	49.86	53.65	52.77	55.56	54.76	55.00
SD	8.88	8.30	8.44	10.11	10.85	10.47

Selecting Texts and Developing Questions

Purpose

The purpose of selecting texts and developing questions was to identify a pool of narrative and informational texts that fit with the schools' reading instructional programs, were at an appropriate level for the struggling readers in the study, and were amenable to the types of questions (Causal and General) under investigation; and to develop Causal and General questions to be used during intervention.

Method

Participants

Participants included the fourth-grade teachers and specialists in the project. We also sought the advice of two 'Discourse Experts,' Dr. Art Graesser of the University of Memphis and Dr. David Rapp of Northwestern University.

Measures

Initial teacher questionnaire. To ensure that we would select texts that would align with schools' instructional programs and the approximate levels that would typically be used with fourth-graders struggling with reading comprehension (but not decoding or reading fluency), we constructed a teacher questionnaire that described the type of reader we planned to target with the intervention, then asked questions about the leveling system the teacher/school used, the specific range of levels at which they expected the struggling readers in our study to be reading during fall, winter, and spring of the school year, the types of reading materials and instructional activities they used, specific books and materials they recommended we use or avoid using, types of text structures they emphasized in reading instruction, and which types of text structures posed particular difficulties for struggling readers.

The questionnaire was sent to all teachers who had agreed to participate in the study, to obtain preliminary information about texts that would fit with their instructional programs and students' reading needs. A copy of the questionnaire can be found in Appendix B.

Coh-Metrix Text Easability Assessor. To identify specific texts, we drafted a detailed description of our text selection process, and sent it to our first 'Discourse Experts,' Dr. Art Graesser, to review and provide feedback. Dr. Graesser generally agreed with our approach, but recommended that we use the web-based *Coh-Metrix Text Easability Assessor* (http://141.225.42.101/cohmetrixgates/) to establish the readability as well as other important dimensions of the text (as opposed to using the full version of *Coh-Metrix*; Graesser, McNamara, Louwerse, & Cai, 2004), as we had originally planned. The *Text Easability Assessor* provides percentile scores on five characteristics of text extracted from a wider range of linguistic features calculated by *Coh-Metrix*: Narrativity, Syntactic Simplicity, Word Concreteness, Referential Cohesion, and Deep Cohesion. The five text easability scores were extracted from a wide range of linguistic features calculated by *Coh-Metrix*, and are recommended for use given that they have been found to account for most of the variance in text across grade levels and text types (Graesser, McNamara, & Kulikowich, 2011). Graesser et al. (2011) define these features as follows:

Narrativity: "The extent to which the text conveys a story, a procedure, or a sequence of episodes of actions and events with animate beings" (p. 228).

Syntactic simplicity: "Scores are higher when sentences have fewer words and simpler, more familiar syntactic structures" (p. 230).

Word concreteness: "Scores are higher when a higher percentage of content words are concrete, are meaningful, and evoke mental images—as opposed to being abstract" (p. 230).

Referential cohesion: "The extent to which explicit content words and ideas in the text are connected with each other as the text unfolds" (p. 230).

Causal cohesion: "Scores are higher to the extent that clauses and sentences in the text are linked with causal and intentional (goal-oriented) connectives" (p. 230).

Text review questionnaire. Once texts were selected, we sent them to the specialists participating in the project, as well as to co-PI van den Broek and Discourse Expert David Rapp, for their review, along with a Text Review Questionnaire (Appendix C). The Text Review Questionnaire asked teachers to indicate whether the texts fit with the types of texts used in their schools' reading curriculum, and asked experts to indicate whether the texts were cohesive enough to allow for comprehension, but also challenging enough for generation of inferences. Both teachers and experts were asked to indicate whether the texts were age appropriate and likely to be interesting to 4th-graders, and sufficiently challenging/demanding for the generation of inferences. They were also asked to give an overall recommendation as to whether the text should be used for the project.

Question development rubric. GRAs were trained to write questions using a question development rubric (Appendix D) that prompted them to read the text; make sure it was coherent, did not require excessive background knowledge, and was age-appropriate; identify the goal structure; identify places where a causal inference would likely be needed; and generate questions designed to prompt such inferences. Guidelines were also provided regarding the wording and formatting of questions.

Procedures

Text development. Texts selected for the intervention had to meet the following criteria:

(1) They had to fit with the types of texts used in participating schools' instructional programs in terms of difficulty level, genre, content, age-appropriateness, and interest level (but did not duplicate specific texts);
(2) They had to be sufficiently difficult but not *too* difficult for students struggling primarily with reading comprehension;
(3) They had to be short enough to be read in no more than 4 sessions; and
(4) They had to have a structure that lent itself to the causal questioning approach to be used in the intervention (e.g., a clear goal with resolution for narrative text; a cause/effect or problem/solution structure for informational text).

In both districts, teachers, reading specialists, and curriculum leaders indicated that they used leveling systems to select appropriate narrative and informational texts for students. District 1 indicated that they primarily used the Fountas and Pinnell Benchmark Assessment System (Fountas & Pinnell, 2006) for leveling text. District 2 reported using a combination of Fountas and Pinnell along with other systems. Based on teachers' questionnaire responses, we determined

that texts leveled in the O-R range were the most likely texts to be used by the type of struggling readers to be included in our study.

Based on all of this feedback, we obtained benchmark passages from the Fountas and Pinnell (2006) system for the O-R range. We did not use these texts for intervention, because they were used for benchmarking purposes in the schools. Rather, we ran these texts through the *Text Easability Assessor* to serve as a benchmark for texts selected for the intervention. Our aim was to ensure that each text fell within the range of readability and percentile scores on the five text features that the benchmark passages did, to ensure that they would represent the text levels typically used in our schools, as well as to ensure that the intervention passages were sufficiently similar to each other on these dimensions.

Finally, we obtained a wide selection of narrative and informational text from participating school reading labs (which included leveled books) and local public libraries. Texts were previewed to make sure they generally met the four criteria listed above; if so, they were transcribed and run through *Text Easability Assessor*. If they fell within the range of proportions of the Fountas and Pinnell benchmark passages, they were selected for the intervention (note that, because the texts were divided into segments, some fell outside of the benchmark ranges on one or two dimensions, but were retained given the overall fit with the benchmarks, and the difficulty in finding texts that perfectly fit all dimensions). We formatted the texts for intervention, and sent representative examples to several of the reading specialists on the project for their review. They also commented directly on the texts (both about the formatting and the content), and completed Text Review Questionnaires.

Question and response development. Four GRAs were trained to develop questions for the intervention texts using the Question Development Rubric (described above). Typically, at least two GRAs developed questions for a text; these were then reviewed by the PI, and at least 8 questions were selected and inserted into the passages. We then sent the passages to co-PI van den Broek for his review, as well as to our second Discourse Expert, Dr. David Rapp. These two experts provided comments using track changes, including edits to existing questions or suggestions for different questions. They also commented on the text itself, and completed Text Review Questionnaires.

The expert feedback was used to make final question selections. Once final questions were selected, they were inserted into the stories. At least two GRAs read each story and questions, and constructed possible responses (including both correct responses and incomplete or incorrect responses that we anticipated students might make). In some cases, the GRAs discovered that responses could not be made based on the text alone, or were ambiguous or awkward, in which case the questions were revised or eliminated and replaced with a new question. The GRAs' responses were used in the construction of scaffolded feedback to be used as part of the intervention (described in more detail under Developing the Intervention).

Results

Table 9 shows the Fountas & Pinnell (2006) benchmark passages, as well as texts selected for the intervention, and the readability and *Text Easability Assessor* dimensions of interest. For the text "Little Singing Frog" used for the Fall Trial Run (described under Developing Intervention below) 8 questions were included for each passage. In the Fall Trial Run, we learned that 8 questions were too many to complete in the intervention time; thus, remaining texts included 5-6 questions.

Table 9. Text Dimensions of Study Passages

			Text Dimension (Target Range)						
Used For	**Type**	**Text-Title**	**Word Count**	**F-K (3.0-4.4)**	**N (>57%)**	**SS (>54%)**	**WC (>68%)**	**RC (<35%)**	**DC (27-80%)**
Benchmark	Narrative	The New Girl (Level O)	406	3.0	90%	78%	81%	35%	55%
Benchmark	Narrative	A Pet for Nate (Level P)	409	4.4	57%	80%	94%	3%	27%
Benchmark	Narrative	Hiding Places (Level Q)	494	4.0	57%	91%	91%	10%	80%
Benchmark	Narrative	The Election (Level R)	530	4.4	78%	54%	68%	11%	51%
Fall Trial Run	Narrative	Little Singing Frog Part 1	459	3.5	80%	63%	98%	42%	45%
Fall Trial Run	Narrative	Little Singing Frog Part 2	584	4.4	75%	48%	98%	36%	49%
Spring Pilot	Narrative	Androcles and the Lion Part 1	250	2.0	43%	99%	94%	29%	47%
Spring Pilot	Narrative	Androcles and the Lion Part 2	347	5.5	81%	73%	95%	83%	89%
Spring Pilot	Narrative	A Different Aladdin Part 1	481	3.5	94%	72%	18%	64%	68%
Spring Pilot	Narrative	A Different Aladdin Part 2	452	3.2	92%	76%	36%	27%	60%
Spring Pilot	Narrative	A Different Aladdin Part 3	443	4.7	88%	48%	87%	22%	74%
Spring Pilot	Narrative	Double-Take Jake Part 1	449	3.6	64%	77%	77%	12%	36%
Spring Pilot	Narrative	Double-Take Jake Part 2	421	3.9	77%	71%	96%	31%	17%
Spring Pilot	Narrative	Double-Take Jake Part 3	439	3.0	93%	64%	82%	57%	50%
Spring Pilot	Narrative	Double-Take Jake Part 4	491	2.3	89%	95%	81%	15%	70%
Spring Pilot	Narrative	The Bremen Town Band Part 1	449	2.9	86%	59%	80%	14%	35%
Spring Pilot	Narrative	The Bremen Town Band Part 2	449	3.7	77%	73%	87%	16%	26%
Spring Pilot	Narrative	The Bremen Town Band Part 3	402	3.5	67%	74%	83%	9%	49%
Spring Pilot	Narrative	The Bremen Town Band Part 4	398	3.7	71%	78%	97%	16%	65%
Spring Pilot	Narrative	The Emperor and the Peasant Boy Part 1	356	3.9	45%	84%	90%	19%	35%
Spring Pilot	Narrative	The Emperor and the Peasant Boy Part 2	372	3.0	82%	75%	35%	14%	10%
Spring Pilot	Narrative	The Emperor and the Peasant Boy Part 3	406	3.7	73%	82%	82%	31%	55%
Spring Pilot	Narrative	The Paper Bag Princess Part 1	338	4.9	67%	86%	82%	62%	44%
Spring Pilot	Narrative	The Paper Bag Princess Part 2	215	3.9	84%	70%	58%	33%	36%
Posttesting	Narrative	Annie Helps a Dolphin	233	3.6	85%	86%	74%	36%	41%
Posttesting	Narrative	The Dragon in the Forest	244	3.3	82%	74%	98%	33%	44%
Posttesting	Narrative	Sam Saves a Squirrel	242	3.0	81%	82%	92%	34%	37%
Posttesting	Narrative	The Spider on the Mountain	235	3.6	80%	74%	99%	34%	45%
Spring Trial	Informational	Jane Goodall Part 1	370	4.8	50%	77%	92%	41%	38%
Spring Trial	Informational	Jane Goodall Part 2	360	2.8	93%	77%	33%	48%	31%
Spring Trial	Informational	Jane Goodall Part 3	342	4.6	64%	93%	90%	17%	87%
Spring Trial	Informational	Olympic Dreams: Part 1	371	5.3	80%	90%	68%	32%	81%
Spring Trial	Informational	Olympic Dreams: Part 2	375	6.5	47%	77%	81%	40%	25%
Spring Trial	Informational	Olympic Dreams: Part 3	319	6.0	62%	93%	81%	20%	64%

Note. F-K = Flesch-Kincaid, N=narrativity, SS = syntactic simplicity, WC = word concreteness, RC = referential cohesion, DC = deep cohesion. Text in red is out of target range (based on benchmark passages).

Developing Intervention

Purpose and Research Questions

The purpose of developing the intervention was to identify one or more feasible and promising questioning approaches to improving struggling readers' comprehension of narrative and informational text. Specific research questions included:

(1) What is the fit and feasibility of the intervention for implementation in school settings?
(2) Do subgroups of struggling readers respond differently to intervention based on the type (Causal vs. General) and timing (online vs. offline) of questioning? In other words, which type and timing of questioning shows promise for the different subgroups?

Below, we first describe the 'final' version of the intervention to provide context for the description of its development. Then, we describe six 'Rounds' of iterative development and refinement that led to the current version of the intervention.

Description of the Final Version of the Intervention: "Making the Right Connections"

Intervention Structure

The research staff (project coordinators, GRAs and URAs) served as tutors who implemented the instruction to participating students in small groups of 3 to 4, with the exception of one group that had five students. The intervention was conducted 3 times per week for 20 to 30 min per session in a small room such as the school's book room, media center, resource room, or in the hallway near the students' classroom.

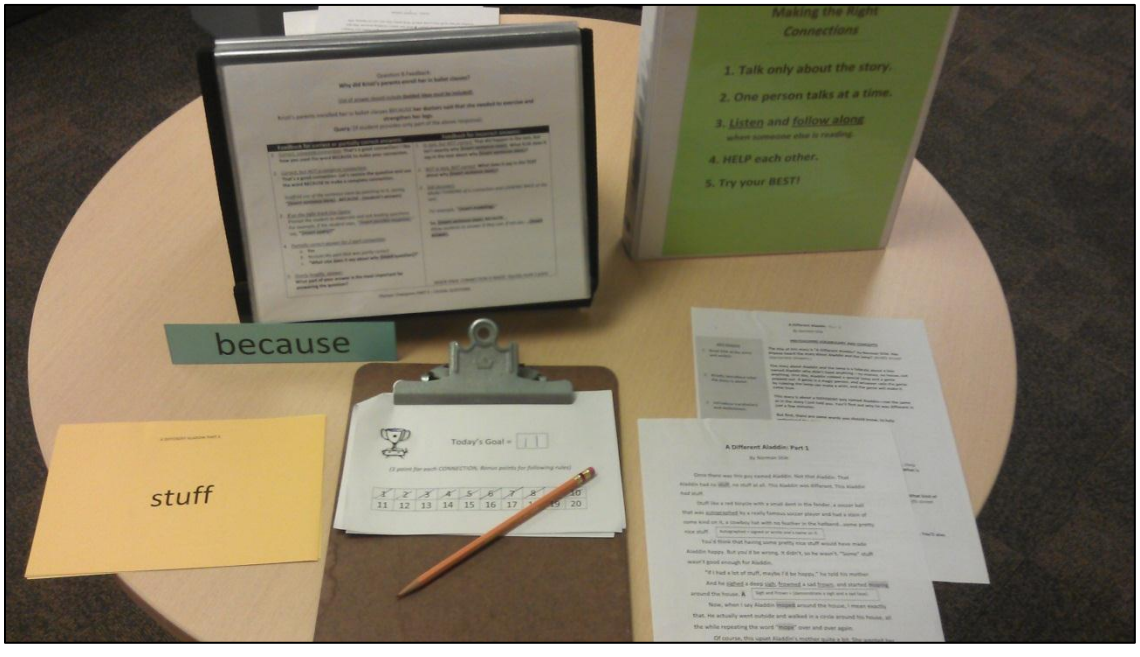
Making the Right Connections was implemented in two Phases (in which students received the Online or Offline version of their intervention in counterbalanced order). For the entire intervention period, 9 groups received Causal questioning and 9 groups received General questioning. Each phase consisted of 10 instructional days. For both phases, Day 1 consisted of an introductory lesson instudents were taught (or reviewed) the purpose of the activity, how to use materials, rules, and how points were awarded. The tutor also modeled answering questions using the following prompt: Look back at the text… Think of a connection… and Connect it! During Days 2 and 3, the groups read a story, and the tutor modeled how to answer the first question, followed by guided practice in which the tutor scaffolded steps needed to answer questions and provided structured feedback. From Day 4 onwards, students read and answered questions as a group. The tutor provided the students with scaffolded feedback based on their answers. (More detail is provided below in the Protocol description).

Intervention Materials

Tutoring materials are displayed in Figure 3. Each student had a two-pocket folder with the purpose of the activity written on the front of the folder as a reminder. Inside the two-pocket folder was a story that was divided into different parts for each intervention session. The story was written in 14 point Calibri font on an A4 size paper and stapled together into a booklet. At various places in the story at the end of the sentence, a highlighted letter was written (see sample in Appendix E). The students also had a blue 8 by 1.5 inch bookmark depending on which condition they belonged to. Students in the Causal condition had a 'because' bookmark. Students in the General condition had a blue 'Connect it!' bookmark.

Tutor materials consisted of a two-pocket folder with a teacher copy of the story, a 'because' bookmark for the Causal condition, or a 'Connect it' bookmark for the General condition. The teacher copy of the story had underlined words with definitions for those words presented in text boxes in the page margin (see sample in Appendix F). The tutor could provide definitions to the group for the underlined words while reading, as needed. The tutor also had a three ring binder with 8.5 by 5 vocabulary cards and scripts for the preteaching component, teacher modeling and ongoing scripts, and rules displayed on the front of the binder, and the Look back-Think-Connect it! Prompt displayed on the back of the binder. There was also a presentation easel with the following materials: rules; the Look back-Think-Connect it prompt card, question card that faced the student, and the teacher feedback card that faced the teacher for each question. The question card had an answer stem to help students answer each question in a complete sentence. There was also a score card on a clipboard, with a box for the tutor to write the goal for the day (the number of points the group should try to receive) with a maximum of 20 points per session.

Figure 3. ***Sample Tutoring Materials for Final Intervention Version***



Intervention Protocol

A typical intervention session consisted of two parts: (1) preteaching with goal setting, and (2) teacher modeling and guided practice. Each intervention session started with goal setting which involved writing the number of points the group would strive to earn on the score card. The preteaching component followed goal setting and was conducted for approximately 5 min. Preteaching involved activating background knowledge and teaching definitions of 3 - 4 words that were central to the story. Activating background knowledge consisted of providing students with a brief preview of the story, or by asking students questions on similar stories they may have read, or experiences they may have had.

The vocabulary words were highlighted on the student and teacher copies of the text. When introducing the vocabulary words, the tutor first showed the vocabulary word written on the index card and said the word aloud. Next, the student read the word aloud with the tutor and then pointed to the same word highlighted in the text. Then, the tutor provided a meaning for the vocabulary word and/or a picture to depict the meaning of the word. The students repeated the

meaning back to the tutor. Next, the tutor asked a question pertaining to the vocabulary word and the student had to respond to the question using the vocabulary word. The above procedure was used to give students several opportunities to actively engage with the vocabulary word. The preteaching component procedure was the same for all questioning conditions.

Causal questioning protocol. For students in the Causal questioning group, after the preteaching component, the tutor implemented either the Causal Online or Offline intervention procedures in Phase I, and vice versa in Phase II. For the Causal Online condition, the questioning occurred during reading, and for the Causal Offline condition, questioning occurred after reading.

In the Causal Online condition, the tutor read the story aloud up to the first highligted letter. The tutor also provided meanings to other unknown words while reading. Next the tutor asked a causal question, displayed on the presentation easel, and prompted students to look back, think of a connection, and share their connection with a peer. After sharing their connection with a peer, the tutor called on one student to 'Connect it,' and the student shared his/her answer. The tutor gave structured feedback based on the student's answer. There were three categories for feedback to answers: correct answer, partially correct answer, and incorrect answer. If the student's answer was correct, the tutor would ask other students if they agreed with the answer and moved on to the next question. If the answer was correct but not answered in a complete sentence, the teacher would prompt the student to answer the question in a complete sentence. If the answer was partially correct, the tutor would prompt the student to elaborate by saying, " That's partly right. What else does it say in the story about why ...(with question stem inserted here)?" The tutor then asked another student to answer the question.

If the answer was incorrect, but did happen in the story the tutor would say, " That did happen in the story, but isn't exactly why … (with question stem inserted here). What else does it say in the story about why…(with question stem inserted here)?" If the answer was still incorrect after two prompts, the tutor would model the answer. If a student gave an answer that was close to the correct response but not correct, the tutor would provide a scaffold by quering the student's response to help the student get to the correct answer.

The group received a point for answering a question, if they worked together to help each other answer the question. Next, the tutor called on another student to read aloud up to the next highlighted letter. After reading, the teacher presented the question and asked students to share their answer with a peer before calling on a student to answer the question. Feedback for each answer was the same as the feedback described in the previous paragraphs. The tutor continued the process of reading and answering questions till all questions for the day were answered or till the last 3 minutes of the session. At the end of each session, if the group met their goal, each student received a sticker to put on their folder beside one of the purpose statements.
The intervention procedures for the Causal Offline condition were exactly the same as those described for the Causal Online condition, with the exception of the timing of question presentation. After the preteaching component, the teacher read up to the first highligted letter, and then called on another student to read up to the next highlighted letter and so forth. The process of reading from one highlighted letter to the next continued till the entire text was read. Next, the teacher presented the first question on the presentation easel and provided feedback to student answers similar to those describe in previous paragraphs.

General questioning protocol. For students in the General questioning group, the tutor implemented either the General Online or General Offline questioning procedure following the preteaching component in Phase I, and vice versa in Phase II. For the General Online condition,

the questioning occurred during reading, and for the General Offline condition, questioning occurred after reading.

In the General Online condition, After the preteaching component, the tutor read aloud up to the first highlighted letter. Next, the tutor pointed to the following question on the presentation easel: "How does this sentence connect to the rest of the story?" The tutor then asked the students to Look back and Think of a connection. Next, students shared their answer with a peer, after which the tutor randomly called on one student to answer the question. The students could use the answer stem provided in the presentation easel to help them answer the question. The answer stem provided was: "This sentence connects to the part in the story where…". Based on the answer the tutor gave feedback. Feedback was given for an answer that made sense in the context of the story, and for an answer that did not make sense in the context of the story.

If the answer made sense, the tutor would say, " Good, your connection makes sense based on what we read in the story. Does everyone think [student's] answer makes sense based on what we've read in the story?". Next, the tutor allowed for a brief discussion and encouraged other students to show how he/she used the story to make the connection. If there was more than one plausible answer, the tutor would say, "Remember, often there is more than one connection that we could make. Did anyone make another connection?"

If the student's answer was related to the story, but did not make sense, the tutor said, "That did happen in the story, but I'm not sure that connection quite makes sense. Let's look for something else in the story that connects to this sentence?" The tutor asked another student to share his or her answer. If the student's answer did not relate to the story the tutor said, "Let's try to make a connection to something that happened in the story. Remember, the last sentence we read was … How does that sentence connect to the rest of the story?" Another student was given an opportunity to answer the question. If the answer was still incorrect after two prompts, the tutor modeled one possible connection. The group received a point for answering the question, and then the tutor called on another student to read aloud till the next letter. The reading and questioning process continued till all questions were answered or up to the final 3 min of the intervention session.

The questioning and feedback procedures for the General Offline condition were exactly the same for the General Online condition, except for the timing of questioning. The tutor read to the first highlighted letter, and then called on another student to read to the next highlighted letter. This process continued for the entire story. Next, the tutor asked the students to point to the first highlighted letter and re-read the sentence before it. The same questioning and feedback procedures described in above were implemented, and the group received a point for each question answered. The session ended by the tutor reviewing whether students met their goals and received a sticker if they did.

Below, we describe the iterative development cycle that led to the version of the intervention described above.

Round 1: In-House Development (September-November, 2011)

The PI and Intervention Project Coordinator produced an initial draft of the intervention script using the outline of the intervention in the grant as well as the existing script from previous research (McMaster et al., 2012). This initial draft underwent several revisions using the following procedures:

(1) We met with co-PI Christine Espin (who visited the U.S. in late September for a conference and was able to meet with us) to walk her through the intervention and obtain her initial thoughts, and revised the script based on her feedback.
(2) We sent the revised draft to co-PI Espin who provided suggestions using tracked changes; these were implemented or discussed/resolved.
(3) We implemented intervention the PI as 'teacher,' GRAs as 'students,' and project coordinators as observers/timekeeper.
(4) GRAs and project coordinators provided open-ended feedback through semi-structured notes and discussion (see Appendix G for feedback form).
(5) We compiled all notes and incorporated this feedback into the draft, maintaining notes of revisions in a summary document.
(6) Two GRAs reviewed the revised scripts and added questions/comments using tracked changes; these changes were implemented or discussed/resolved.
(7) All research staff met and reviewed the revised draft of script; GRAs provided verbal feedback during this meeting and we incorporated minor revisions based on this feedback.
(8) We sent the revised draft to co-PIs van den Broek and Espin, and incorporated their minor feedback (using track changes).
(9) All research staff participated in a 'dress rehearsal' of the intervention as it would be demonstrated to teachers during Focus groups.
(10) Minor problems were identified and addressed during this dress rehearsal and incorporated into a final draft.

Round 2: Focus Groups (November, 2011)

The purpose of Round 2 was to obtain feedback from teachers on the intervention's *fit* with their schools' instructional programs and students' needs and *feasibility* to be implemented within the schools' instructional structures and schedules.

Participants

All participating teachers were invited to attend one of two half-day focus group sessions. In Session 1, 8 classroom teachers and 2 specialists attended. In Session 2, 11 classroom teachers and 5 specialists attended. Two 'make-up' sessions were held for 2 classroom teachers and 2 specialists. Two teachers did not attend, as their building principal preferred only 2 teachers attend as representatives.

Measures

The Focus Group Questionnaire, described under procedures, is in Appendix H.

Procedures

Both focus group sessions followed a similar format. First, project staff provided an overview and rationale for the intervention, as well as a timeline of project activities and description of how students were identified as eligible. Then, project staff demonstrated the intervention, while teachers followed along on 'student' copies of the story used during demonstration. Demonstration was divided into four parts: (1) Introduction, (2) Pre-Teaching, (3)

Teacher Modeling and Practice, and (4) Ongoing Instruction. Teachers were provided a list of open-ended questions related to each part of the demonstration. After each part of the demonstration, project staff sat with groups of teachers (approximately 8-10 teachers per group) and asked each question, following a "Facilitator Version" of the Focus Group Questionnaire. This version prompted the researchers to review responses to each question and prompt additional thoughts and comments. At least two researchers took detailed notes on the Facilitator Version forms, and the session was audio-recorded. Teachers were not required to provide written responses, but were encouraged to provide additional written feedback if they did not feel comfortable stating ideas aloud in the group. All feedback from teachers was entered into an excel spreadsheet.

Following the focus group sessions, all research staff who had attended the sessions independently wrote and then discussed what they thought emerged as the major themes of the teachers' feedback (based on their notes). There was very good convergence across research staff members as to what the major themes were. The PI took notes on this discussion, then compiled all notes into the excel spreadsheet with feedback from teachers. The themes identified by the research staff were used to categorize all feedback. We then identified aspects that could be incorporated into the intervention for trial runs (we found that nearly all feedback could be addressed in revisions).

Two additional mini-focus groups were held with teachers who could not attend the main focus group days: one with two 4th-grade teachers from a school in District 1, and one with two specialists from a school in District 2. The PI presented the same overview/rationale as was presented in the larger groups. She then demonstrated the intervention, focusing on the areas that had been revised based on previous focus groups' feedback. The teachers provided additional feedback, which was added to the excel spreadsheet and used in the final revision process. We incorporated all focus group feedback into intervention scripts and materials.

Note that all of the above activities were completed with an "Online Causal" version of the script. We waited to make the other versions until as much feedback was incorporated as possible, to avoid making revisions to four different versions of the script. Once revisions were completed, we altered the Online Causal Version into Offline Causal, Online General, and Offline General versions.

Once all versions were completed, the GRAs were given scripts, stories, and other materials, and revisions were highlighted and discussed. The GRAs were given time to practice all scripts with each other, and they sent additional minor corrections identified during this practice time. Finally, the GRAs conducted a dry run of the intervention, with each GRA taking a turn as the interventionist and the other GRAs acting as students. The PI and project coordinators observed, provided feedback, and made note of final minor corrections to be made. Finally, the PI drafted a fidelity tool to be used with each version of the intervention during the Fall Trial Run. This tool was revised during the Fall Trial Run (Round 3), and the final tool is described in detail under *Measures* for the Spring Pilot (Round 4).

Results

Table 10 shows the categories of feedback and examples of revisions made based on focus group feedback.

Table 10. ***Focus Group Feedback Revisions***

Category	Revisions Made
General delivery of intervention	• Used University of Minnesota folders for student materials (to motivate students) • Added language to emphasize that students could 'look back' at the whole story • Put the purpose of the activity on the student folders
Introduction	• Emphasized importance of working as a team • Emphasized purpose and importance of becoming a better reader • Clarified to students why they were selected • Provided examples of questions • Added discussion/rationale for rules • Explicitly discussed how use of term "making connections" relates to schools' use of term
Pre-teaching	• Used pictures to convey concepts • Challenged students to use vocabulary terms in their answers
Modeling	• Used question stems to scaffold answer generation • Made more clear that answers can be found in any part of the previously-read text
Ongoing instruction	• Made a more clear ending to lesson • Expanded on feedback given for partially correct/incorrect answers

Round 3: Fall Trial Run (December, 2011)

The purpose of the trial run was to determine the feasibility and flow of the *structure* of the intervention. In other words, we examined whether the intervention was operating as intended.

Participants

The trial run was conducted with a subset of students identified as eligible for intervention (n = 50). These students were selected based on availability of both students and tutors. Students were placed into 14 groups of 3-4 students based on their school schedules.

Measure

Intervention Logs. Tutors (three GRAs and the Intervention Project Coordinator) implemented maintained Intervention Logs of each session that included the total number of minutes of instruction, student attendance, and the number of questions answered. They also recorded ratings regarding the feasibility and appropriateness of the intervention components after each session. Ratings were on a Likert-type scale of 1 to 4 (1 = disagree to 4 = agree, or 1 = infeasible to 4 = feasible). Tutors were asked to note specific issues/problems and suggestions for improvement for items rated lower than 4. Tutors brought their intervention logs to lab meetings to facilitate discussions regarding whether the intervention was operating as intended or additional revisions were needed. Mean ratings were obtained for each component, and open-

ended feedback was summarized and used for further revision to the intervention. The Fall Trial Intervention Log is found in Appendix I.

Procedures

The trial run was conducted for two weeks in December. In Week 1, 27 students received the Causal Questioning version of the intervention, using narrative text; 16 received the online version, and 11 received the offline version. In Week 2, 23 new students received the General Questioning version of the intervention, using narrative text; 10 received the online version, and 13 received the offline version.

During the trial run, the PI and one other researcher observed each tutor at least twice (some together, some separately), ensuring that each combination of questioning type and timing was observed. The two researchers discussed and refined the Fidelity Form following these observations to ensure that it would capture critical elements of the intervention, as well as important differences across each version.

At the end of each week, all research staff met to discuss the intervention, using the Intervention Logs to guide discussion. Detailed notes were taken during these meetings, and Intervention Log data, including ratings and comments, were summarized. Each category of feedback was reviewed systematically, discussed among the PI, co-PIs, and project coordinators during Skype meetings, and incorporated into revisions of the intervention scripts and materials. All revisions were reviewed by co-PIs van den Broek and Espin.

Results

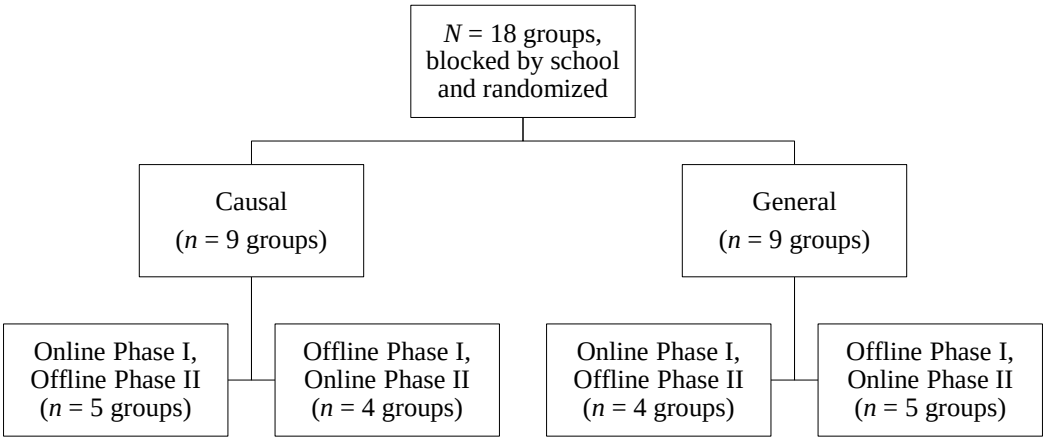
The most significant revisions made based on the trial runs were (a) improving materials so that they were less cumbersome/easier to use, (b) reducing the amount of teacher talk so that students would have more opportunities to respond and could complete the lesson, (c) adding more modeling and feedback, and (d) reducing the number of questions from 8 to 6 so that students could complete the story and the lesson in the allotted time.

Round 4: Spring Pilot (Late January through April, 2012)

Purpose and Design

The purpose of the Spring Pilot was to obtain preliminary data regarding the *type* (Causal vs. General) and *timing* (online vs. offline) of questioning showed promise for subgroups of struggling comprehenders. We used a quasi-experimental, randomized mixed design with one between-groups (TYPE: causal vs. general), one within-subjects (TIMING: online vs. offline) variables, as illustrated in Figure 4.

Figure 4. ***Assignment to Question Type (Causal, General) and Timing (Online, Offline)***



Participants

All students identified as eligible for intervention participated (n = 62). Because the intervention was conceived of as a small-group intervention, students were placed in groups of 3-4 (and one group of 5) within their schools (within or across classrooms, as scheduling dictated). Groups (n = 18) were assigned randomly to Causal vs. General within school (when possible) or within 2 schools with similar demographic make-ups (when within school was not possible). Causal and General groups were then assigned randomly to either receive the Online or Offline version in Phase I (Weeks 1-3), and the opposite version in Phase II (Weeks 4-6).

Demographic and screening data for students in the two groups are provided in Table 11, along with chi-square and F-statistics. There were no statistically significant differences between the two groups in terms of demographic or screening (pre-treatment) performance.

Table 11. ***Comparison of Demographic and Screening Data by Questioning Group***

	Causal (n = 30)		General (n = 32)			
	N	(%)	N	(%)	χ^2	p-value
Sex (Male)	16	53%	11	34%	2.26	.132
Race/ethnicity					4.076	.396
African American	12	40%	9	28%		
Asian	1	3%	2	6%		
White	8	27%	9	28%		
Hispanic	7	23%	12	38%		
Other	2	7%	0	0%		

Table 11 ***continued.***

	Causal (n = 30)		General (n = 32)			
	N	(%)	N	(%)	χ^2	p-value
Receive Free/Reduced Lunch	20	74%	19	66%	0.484	.487
Received Special Education	4	13%	4	13%	0.01	.922
Autism	0	0%	1	3%		
Speech/Lang Impairment	1	3%	2	6%		
Learning Disabilities	0	0%	1	3%		
Other Health Disability	1	3%	0	0%		
SNAP	2	7%	0	0%		
Received ELL Services	8	27%	12	38%	0.832	.362
Received Reading Support	10	35%	16	50%	1.498	.221
	Mean	(*SD*)	Mean	(*SD*)	F(df)	p-value
Age (at start of school year)	9.43	0.28	9.42	0.32	0.02 (1, 60)	.903
Screening Scores						
CBM Maze	11.45	2.73	11.20	2.94	0.12 (1, 60)	.733
District 1 Test Score	189.53	8.73	190.83	12.39	0.11 (1, 27)	.742
District 2 Test Score	194.77	8.73	193.05	13.28	0.17 (1, 30)	.686
CBM Oral Reading	118.05	18.10	111.23	17.25	2.31 (1, 60)	.134

Note. SNAP = Student needing alternative programming; ELL = English Language Learner.

Measures

Several types of measures were used during the Spring Pilot, including measures of *fit and feasibility*, *fidelity, social validity* and *promise* of the intervention.

Fit and feasibility. Fit and feasibility feedback was solicited from both tutors who implemented the intervention and from reading specialists who observed it. Tutors maintained logs of each session that included the total number of minutes of instruction, student attendance, and the number of questions answered. These data were compiled for both group and individual dosage information (total sessions completed, total minutes completed, and total questions answered). Space was also provided for tutors to note any special issues related to individual student attention and behavior, as well as an special issues related to the session such as what went well, didn't go well, and suggestions for improvement. A Spring Pilot Intervention Log is in Appendix J.

In addition to the tutoring logs, four times across the Spring Pilot, tutors completed an Intervention Rating form based on their experiences implementing the intervention. The Intervention Rating form asked tutors to rate the extent to which they agreed with statements about the intervention purpose, rules, and points (e.g., 'Students understand the purpose of each activity,' 'Students follow the rules,' 'Students like the point system'); the extent to which they agreed with statements about the pre-teaching activities (e.g., 'Activated student background

knowledge about the story,' 'Provided clear definitions of key vocabulary') and modeling, practice, and ongoing instruction (e.g., 'Provided sufficient opportunities for all students to read,' 'Provided sufficient feedback to students to make good connections'); and the feasibility of aspects of the intervention (e.g., 'Implementing all lesson components,' 'Following script,' 'Using materials.') Ratings were on a 1-4 Likert-type scale (1 = disagree to 4 = agree or 1 = infeasible to 4 = feasible). Intervention Rating forms were completed at the beginning of lab meetings (held at the end of each week of the Spring Pilot) and used as the basis for group discussion about the intervention. The Intervention Rating form is provided in Appendix K.

Also during the Spring Pilot, four of the participating specialists observed the intervention and provided feedback on an Intervention Questionnaire. This questionnaire asked the specialists to *'rate each core intervention component on its fit with instructional programming and student needs in your school, as well as its feasibility for implementation in school-based settings'* and to elaborate on their responses with comments and suggestions. Items to be rated included the interventionist (adult tutor), group size, duration and frequency, text, pre-teaching activities, questioning type and timing, number of questions, response mode, feedback, and motivational system. Items were rated on a 1-4 Likert-type scale, and included both fit (1 = not a good fit; 4 = very good fit) and feasibility (1 = not feasible; 4 = highly feasible). The Intervention Questionnaire is provided in Appendix L.

Fidelity. Fidelity forms for each version of the intervention (Online Causal, Offline Causal, Online General, and Offline General) drafted for use during the Fall Trial Run (Round 3) were revised for use during the Spring Pilot (Round 4). Fidelity Forms were identical except for the key features that distinguished each version (type and timing of questions). Each form included several items under each of the following sections for Set-up (3 items); Introduction (3 items); Pre-teaching (4 items); Modeling (6 items); Ongoing Instruction (7 items); and Delivery, Motivation, and Engagement (4 items). Some items were rated as 0 or 1 (e.g., 'student materials are ready'), whereas others were rated on a scale of 0 to 3 (0 = No, 1 = Somewhat, 2 = Mostly, 3 = All/Yes; e.g., 'Teacher is familiar with script;' 'Students appear to be engaged for most of the session'). Also, because each story had up to six questions (labeled A-F), items under Ongoing Instruction were rated for each question (e.g., for 'Teacher prompts students to LOOK, THINK CONNECT,' the rater would circle the letter for each question for which this behavior was observed). A sample final version of the Fidelity Form (Causal Online version) is in Appendix M.

Social validity. Social validity feedback was solicited from students who participated in the intervention. Students completed a questionnaire, which was read aloud to them individually during posttesting at the end of the Spring Pilot. Questions asked students to rate the helpfulness of components of the intervention (e.g., 'How helpful was it to answer questions and make connections to understand the stories we read during Making the Right Connections?') and how much they liked components of the intervention (e.g., 'How did you feel about setting a goal for the number of points to earn each session). Students also rated how well they liked each of the stories they read during the intervention. All items were rated on a 1-4 Likert-type scale (1 = very unhelpful to 4 = very helpful, or 1 = I disliked a lot to 4 = I liked a lot). Thumbs-up/thumbs down pictures were paired with the ratings to make sure it was clear what the student was being asked to do. Following the ratings, students were asked three open-ended questions: (1) 'When you are reading a story, what are some things you can do to help you understand what you are reading?' (2) 'How could we make the Making the Right Connections activity more fun?' (3)

'Would you recommend the Making the Right Connections activity to other kids like you? Why or why not?' The student Social Validity questionnaire is in Appendix N.

Tutors also completed a questionnaire after the Spring Pilot was completed. Questions asked tutors to rate the helpfulness of components of the intervention (e.g., 'How helpful was answering questions and making connections in helping your students understand the stories they read during Making the Right Connections?') and how much they liked components of the intervention (e.g., 'If you implemented the CAUSAL condition, how did you feel about asking your students to answer causal questions?'). These items were rated on a 1-4 Likert-type scale (1 = very unhelpful to 4 = very helpful, or 1 = I disliked a lot to 4 = I liked a lot). Tutors were encouraged to elaborate on their responses, including suggestions to improve the intervention. Tutors also rated the appropriateness of each of the stories in terms of age appropriateness, interest, and level of challenge (not too easy, not too difficult) on a 1-4 Likert-type scale (1 = very inappropriate to 4 = very appropriate). Finally, tutors were asked, 'If you could change anything about the intervention, what would you change and how would you change it?' The tutor Social Validity questionnaire is in Appendix O.

Promise. The Spring Pilot study was conducted in two Phases (as described in Procedures). After each phase, posttests were given. Below, we describe posttest texts and dependent variables.

***Posttest texts*.** Students read one of two pairs of texts at each posttest. These texts were similar in format and readability level to those read during the intervention. Each pair of texts included a 'near transfer' text and a 'far transfer text.' Near transfer texts were administered in a similar format to the format used during training (with the same types and timings of questions asked), and far transfer texts were administered without the questioning. Including near- and far-transfer texts allowed us to examine immediate effects of the questioning intervention, as well as whether intervention effects transferred when the intervention was no longer provided.
The two near-transfer texts, "Annie Helps a Dolphin" and "Sam Saves a Squirrel," were designed to be parallel to each other, in that they followed a very similar story structure and general plot with similar goals, sub-goals, and resolutions, but with different characters and specific scenarios. The two far-transfer texts, "The Dragon in the Forest" and "The Spider on the Mountain," were also designed to be parallel to each other in the same way. "Annie Helps a Dolphin" was paired with "The Dragon in the Forest" (Pair A) and "Sam Saves a Squirrel" was paired with "The Spider on the Mountain" (Pair B). All texts were similar to each other and to stories used in the intervention in terms of readability and the five Text Easability features (shown in Table 8). Pairs of texts were counterbalanced such that half of the students in each condition (Question Type and Timing) were administered Pair A after Spring Pilot Phase 1 and Pair B after Spring Pilot Phase 2, and vice versa. Table 12 shows the counterbalanced order of text pairs administered by condition in Phases 1 and 2. Examiner versions of the Causal and General posttest protocols are in Appendix P.

Table 12. ***Counterbalancing of Posttest Text Pairs by Condition and Phase***

Question Type	Phase 1 Question Timing	Phase 2 Question Timing	*n*	Phase 1 Posttest Text	Phase 2 Posttest Text
Causal (*n* = 30)	Offline	Online	6	Pair A	Pair B
			6	Pair B	Pair A
	Online	Offline	9	Pair A	Pair B
			9	Pair B	Pair A
General (*n* = 32)	Offline	Online	8	Pair A	Pair B
			8	Pair B	Pair A
	Online	Offline	8	Pair A	Pair B
			8	Pair B	Pair A

Note. Pair A = Annie Helps a Dolphin (near transfer) and The Dragon in the Forest (far transfer); Pair B = Sam Saves a Squirrel (near transfer) and The Spider on the Mountain (far transfer).

***Dependent variables*.** The following dependent variables were obtained from the near and far transfer texts:

(1) Responses to questions using intervention approach (near transfer only). The student read the post-intervention text aloud, responding to 5 questions in the same way as during intervention (Causal or General, Online or Offline). The score was the number of correct responses.

(2) Number of prompts needed to arrive at final answer (near transfer only). When the student responded to questions, the examiner followed the following prompting system:

a. Correct on first try: no prompt.
b. Partially correct: examiner stated that the answer was partially correct and prompted a more complete answer.
c. Incorrect on first try: examiner stated that the answer either made sense but was not in the story, or did not quite make sense based on what was read so far, and prompted another answer.
d. Incorrect on second try: examiner stated that the answer still did not make sense based on the story, and repeated the question.
e. Never correct: examiner terminated the prompting and said, “OK, let’s move on.”

To decide which prompt to provide, the examiner followed a guide (for either the Causal or General questioning condition) that outlined what should be included in a complete, correct answer (for the Causal condition), which prompts to use depending on the student's answer, and a script for each prompt. Guides for the Annie and Sam stories are provided in Appendix Q. The examiner circled each level of prompting used for each scoring sheet (see Examiner Protocols in Appendix P). Student protocols were later coded and scored for the number of correct answers with no prompts, the number of correct answers with partial prompts, the number of correct answers with 1 or 2 full prompts, and the number of answers that were never correct despite prompting.

(3) Verbal recalls (near and far transfer). After reading and responding to questions (for the near transfer text) and after reading only (for the far transfer text), the student was asked to recall the text just read. Specifically, the students were prompted, *"Now, let's pretend that I am a friend from your class, and I want to know all about what happened in the story. Please tell me as much of the story as you can remember. What happened first?"* The student was additionally prompted, *"What happened next?"* until the student finished retelling. If the student did not tell the very end of the story, the student was finally prompted, *"Is there anything else about the story that you'd like to tell me?"*

Students' responses during the recall of each text were audio recorded, transcribed, and parsed into thought units based on the original text. Each recalled thought unit was compared to the parsed story units in the original text and coded using a scheme adapted from Kendeou and van den Broek (2005). The coding included the original story unit number to which it matched (if any), and the extent to which it matched or was constrained by the important causal connections in the story. The following five codes were recorded:

- *Conservative* (directly matches a story unit and is a literal or nearly literal rendering of the original story unit)
- *Liberal* (directly matches a story unit and captures the gist of the story unit (the essence of its meaning). Paraphrasing may be present.)
- *Matched but incorrect* (directly matches a story unit, but is significantly incorrect. MI is reserved for absolutely and importantly contradictory interpretations, not just an erroneous detail of the story)
- *No match consistent* (cannot be matched directly with the gist meaning of any one (or two adjacent) story unit, but is valid, consistent with (constrained by) the information in the text, or plausible.)
- *No match inconsistent* (invalid, inconsistent (unconstrained by) the information in the text. The thought unit should be flat-out incorrect to be NMI)

The number of matches to *highly connected* original story units was also counted. Highly connected story units were those that either had at least six or seven (depending on the story) connections to other story units, or stated a goal or attainment of a goal in the story. The recalls were also coded for *order of recall*, which was the correlation of the sequence of recalled thought units with sequence of the original text story units.

It should be noted that the *matched but incorrect* code was new. The research team believed it was useful to distinguish between recalls that accurately reflected the story

unit (*conservative* or *liberal*) from those that matched a story unit but reflected a significant misunderstanding. The *matched but incorrect* code was applied cautiously 17 times for thought units that directly contradicted the story unit in an important way.

(4) *Words read correctly in one min* (near and far transfer). After completing the verbal recall of the near or far transfer text, students re-read the text aloud for one min, following the same procedures used during the CBM Oral Reading task described above. WRC were recorded.

Procedures

Tutor training. The two project coordinators, seven GRAs and three URAs served as tutors during the Spring Pilot. Before intervention began, tutors participated in a two-hour training session. During this session, we (1) reviewed the purpose of the study and research questions, walked through all of the materials, and discussed key considerations for implementation, such as ensuring multiple opportunities to respond, maintaining student engagement, providing appropriate modeling and feedback, and managing behavior. The intervention was demonstrated, with the PI serving as tutor and undergraduate volunteers serving as students. Tutors were encouraged to ask questions, and were given a quiz to self-check their learning. They were then given approximately one week to practice the intervention with each other. When they were comfortable with the intervention, they met with the PI or a project coordinator to 'check out'—in other words, they needed to be able to implement all components of the intervention and use the materials accurately (at least 90% based on the revised fidelity tool) and fluently (e.g., making eye-contact, using a conversational tone of voice, not over-relying on scripts).

Intervention implementation. Implementation began in late January and lasted through the end of March. Each tutor began Phase I with a generic introductory lesson, followed by 9 lessons using their group's assigned *type* (Causal or General) and *timing* (Online or Offline) of questioning. Phase I was followed by individual posttesting sessions, which took approximately one week (no intervention was implemented at this time). When posttesting was complete, the tutor implemented Phase II, which consisted of 9 lessons. Phase II lessons were delivered using the same *type* of questioning as in Phase I, but the opposite *timing.*

Fidelity observations. During the Spring Pilot, two researchers observed 4 out of 18 intervention groups (22%) together to obtain interrater agreement. Interrater agreement, calculated as the number of agreements divided by the number of agreements plus disagreements, ranged from 89.9% to 98.1% (mean = 93.58%). After establishing interrater agreement, the two researchers observed each tutor at least once during Phase I and again during Phase II.

Recall coding training and reliability. Training on verbal recall coding began after the co-PIs van den Broek and Espin (who served as recall coding experts on the study) parsed the four posttest narrative texts into story units, and identified the causal connections between the story units. (See Appendix R for parsed stories with identification of antecedent and consequent connections, and highly connected story units.)

Recall coding guidelines from previous studies were revised for the current study. During training and practice, the team built glossaries of exemplary coded thought units (See

Appendix S for coding guidelines and glossary). The PI and the project coordinators practiced coding three student recall protocols and consulted with co-PIs van den Broek and Espin.

The glossaries and the procedure document were the basis of GRA training on recall coding. Drafting the procedures, conducting training of GRA coders and practicing by the whole team took approximately three weeks. The team discussed and practiced applying the coding procedure as a group. Then, pairs of GRAs practiced coding three protocols jointly with a project coordinator. Project coordinators also practiced coding as pairs. Questions were discussed and resolved at team meetings by the PI or by the coding experts; examples were added to glossaries. Coding pairs then coded four protocols jointly and the PI performed IRA for each coding pair. Although IRA for pairs was initially low, when pairs achieved at least 75% IRA on codes and on story unit numbers, the GRAs in the pair were approved to code independently.

After coding reliability was demonstrated, pairs coded one protocol in common for every two protocols coded independently. The PI made IRA checks on independent and consensus-coded protocols. 10% of protocols for each pair were selected randomly for IRA. If the protocol selected for IRA had fewer than six lines of transcribed recall, it was deemed too short for useful IRA and the next subject number was reviewed. The protocol with six or more lines of transcribed recall was used for IRA coding. Random selection of 10% of each set of protocols assigned to individuals or pairs was repeated for IRA until all coding was complete.

The IRAs on recall coding were calculated for agreement on codes and agreement on story unit numbers of *conservative* or *liberal* matches. The table below shows IRA ranges and averages for pairs and for the third-party (the PI) with individuals and pairs. Note that until the third-party IRA with an individual or a pair was at least 75%, PI coding and IRA checking continued. The average IRA across all consensus protocols was 75% on codes and 88% on story unit numbers.

During the four weeks of coding, the PI, PCs, and GRAs met weekly to practice coding of at least one recall protocol to maintain calibration. The protocols that had been practice coded during the training period were recoded in the routine manner toward the end of the coding period.

Data coding and entry. After both phases of intervention were complete, all data were transcribed (as needed), coded, and entered into an excel spreadsheet. All relevant codes were counted, and counts were double-checked by a second coder. All data were entered by pairs of GRAs, who entered all data twice, into two separate excel worksheets. For the first entry, one GRA called out the data and the other entered the data. For the second entry, the GRAs switched roles. The two worksheets were then compared using the logical test formula in Microsoft excel, and all discrepancies were resolved. Excel sheets were converted to SPSS 19.0 for analysis.

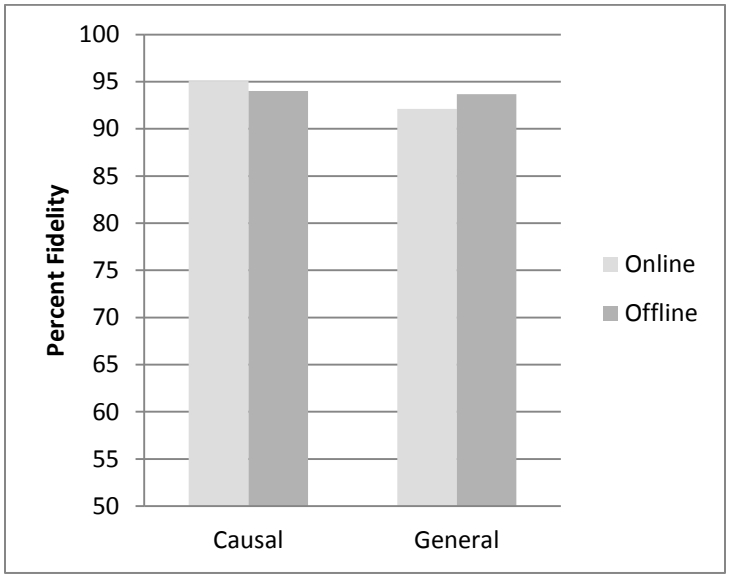
Results

Research Question 1: What is the fit and feasibility of the intervention for implementation in school settings?

Fidelity. Fidelity observations were conducted to determine whether the tutors implemented the intervention as intended. Whereas this information is important for interpreting student response to the intervention, it is also important to know whether the tutors were able to implement the intervention based on the training and support that they received to implement it. Figure 5 shows the mean proportion of fidelity with which the intervention was implemented

across type (Causal, General) and timing (Online, Offline) conditions. Generally, tutors implemented all versions of the intervention with high levels of fidelity, suggesting it was feasible to implement accurately.

Figure 5. ***Fidelity Percentages by Type and Timing of Questioning***

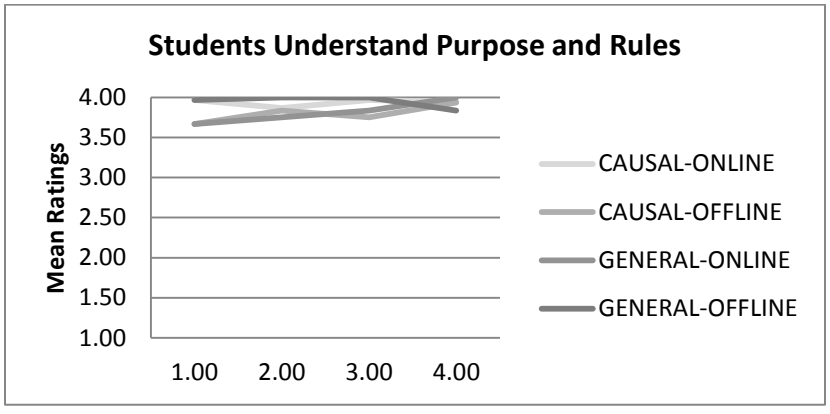


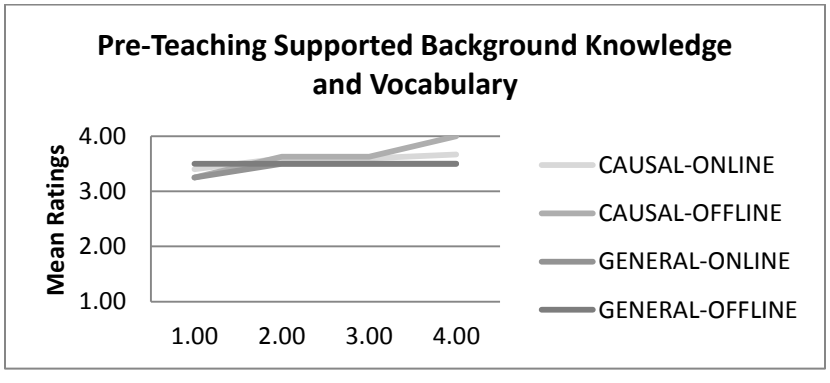
Tutor ratings. Four times during the intervention, tutors completed Intervention Rating forms to indicate their impressions of the feasibility of the intervention. **Figure 6** shows results of their ratings across the four sessions. Generally, they found the intervention to be feasible to implement, with most mean ratings between 3.5 and 4.0. Their view of the feasibility of the pre-teaching activities improved over time, most likely due at least in part to the fact that we made minor improvements to the vocabulary (e.g., adding simple definitions and pictures) over time.

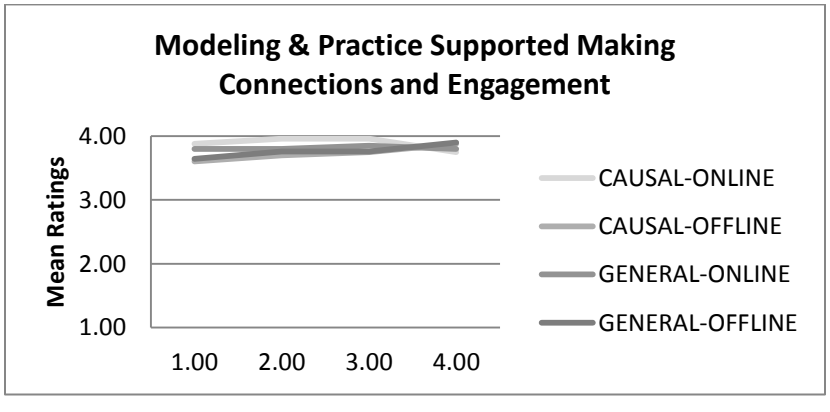
Specialist ratings. Four specialists observed implementation of the intervention, and completed ratings of the fit and feasibility of the intervention components. They also provided open-ended feedback to elaborate on their observations. Table 13 provides a summary of their ratings. Overall, specialists rated the intervention as having a good to a very good fit. They particularly liked the pre-teaching component. One specialist thought that the number of questions was too high for some of the struggling readers. Most of the specialists thought that the motivation system (goal-setting, points, and stickers) did not fit well with their school's philosophical approach to motivating students.

Social validity. Students rated the social validity of the intervention. Figure 7 shows results of student ratings of how much they liked and found the intervention to be helpful. On average, they rated working as a team, setting goals, and earning points and stickers between 3.5 and 4, indicating that they liked those aspects of the intervention. Average ratings for answering questions and making connection were between 3.0 and 3.5, indicating that they found those aspects of the intervention to be at least somewhat helpful. There appeared to be a slight preference for answering questions *after*, rather than *during*, reading.

Figure 6. ***Tutor Intervention Ratings***







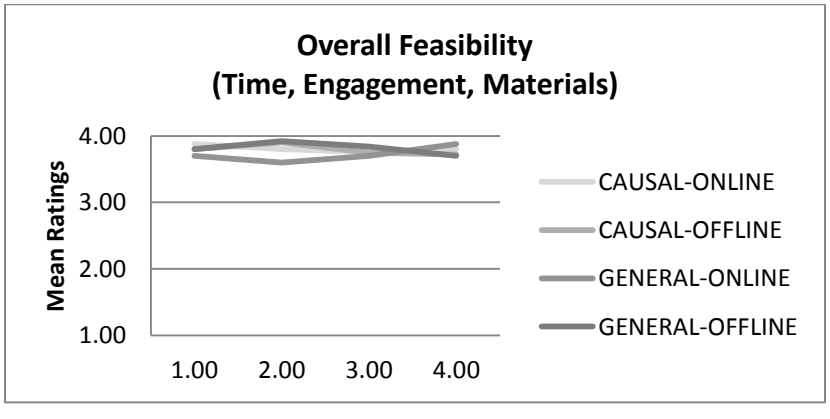
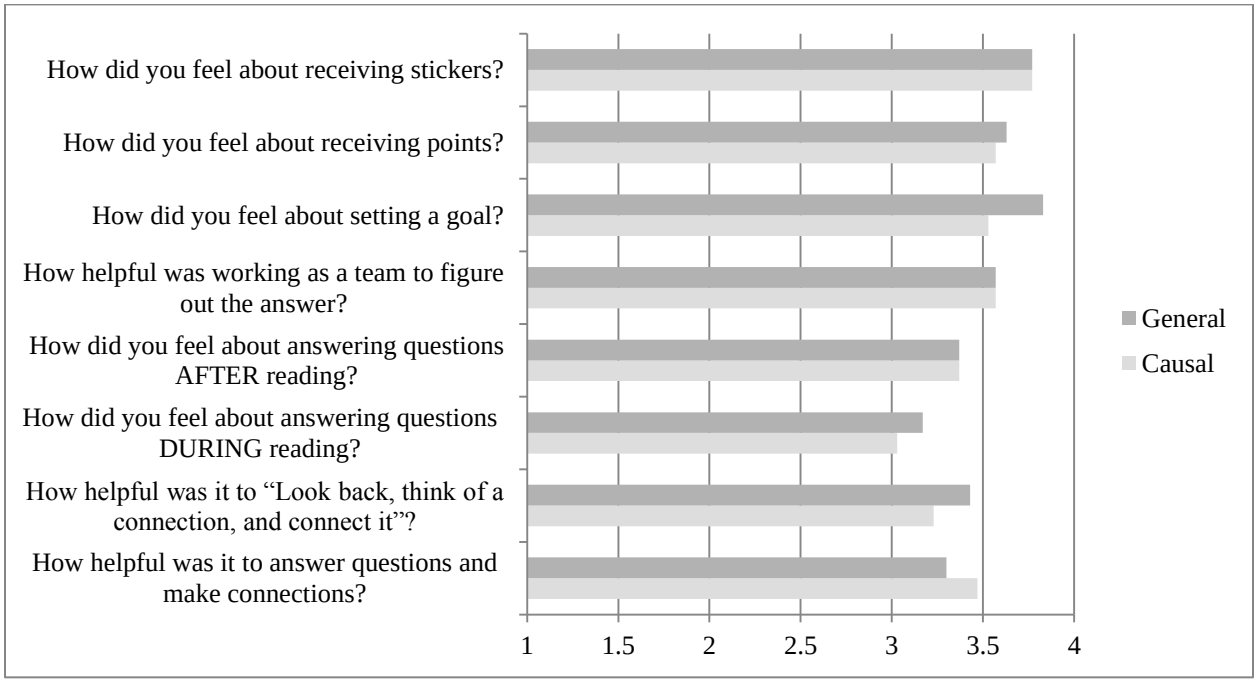


Table 13. ***Specialists' Ratings of Intervention Fit and Feasibility***

Core Intervention Component	FIT 1 = not a good fit; 4 = very good fit	FEASIBILITY 1 = not feasible; 4 = highly feasible
Interventionist: Adult tutor	3.40	3.20
Group size: 4 students	3.80	3.50
Session Duration: 30 min	3.67	3.67
Frequency: 3 times per week	3.67	3.50
Text: Narrative, short stories	3.40	3.40
Pre-teaching (preview key vocab/concepts)	4.00	4.00
Questioning Approach (Causal)	3.25	4.00
Questioning Approach (General)	3.67	3.67
Timing of Questioning (Online)	3.13	3.00
Timing of Questioning (Offline)	3.50	3.50
Number of Questions: Approximately 6 per text	2.80	3.00
Response mode: Partner + individual sharing	3.17	3.33
Feedback	3.50	3.50
Motivational system	2.88	2.75

Figure 7. ***Student Ratings***



Research Question 2: Do subgroups of struggling readers respond differently to intervention based on the type (Causal vs. General) and timing (online vs. offline) of questioning?

Preliminary analyses have been conducted to answer this question. Specifically, repeated-measures multivariate analyses of variance (RM-MANOVA) were conducted with one within-groups variable (timing of questioning) and two between-groups variables (type of question and subgroup) for near- and far-transfer recall and CBM oral reading outcomes. Due to the exploratory nature of this study, preliminary analyses have been conducted at the student level to maximize power. Also, given that there were only two students in the Questioning subgroup, they were excluded from statistical analyses at this time. Their data will be examined using a case study approach to better understand whether the 'questioning' designation is a meaningful one, based on whether they differ from other students on descriptive measures or their response to the intervention. Means and *SD*s for our primary dependent variables are provided in Table 14. Below, we summarize statistical results for the recall near- and far-transfer variables, and oral reading (wrc) near- and far-transfer variables.

For all variables (near- and far-transfer recall and CBM oral reading, there were no statistically significant main effects for questioning type or subgroup, and no type by subgroup interactions. There were also no statistically significant main effects of timing, and interactions among question timing, type, or subgroup. Because the purpose of this project is developmental, we plan to further examine the comparisons in terms of effect sizes, to determine if there are any potentially meaningful patterns in the data.

Figures 8 and 9 are examples of graphs that will help us identify possible patterns of interest (we selected highly-connected story units recalled as this variable is intended to best reflect students' construction of a coherent representation of text). In general, while not reliable, there does seem to be a trend favoring the Causal version of the questioning intervention (more pronounced in near transfer recall), and NOT favoring the General Offline version. If further examination of the data is consistent with this interpretation (including dependent variables that still need to be examined: number of questions answered correctly and numbers of prompts given), it is possible that we would select the Causal version for continued development. However, at this point, that conclusion is very tentative.

Table 14. ***Means and*** **SDs** ***for Recall and CBM Oral Reading Measures***

	Causal						General					
	Online			Offline			Online			Offline		
	Paraphrasers (*n* = 7)	Elaborators (*n* = 23)	Total (*n* = 30)	Paraphrasers (*n* = 7)	Elaborators (*n* = 23)	Total (*n* = 30)	Paraphrasers (*n* = 9)	Elaborators (*n* = 19)	Total (*n* = 28)	Paraphrasers (*n* = 9)	Elaborators (*n* = 19)	Total (*n* = 28)
Recall												
Near Transfer												
Conservative (C)												
Mean	0.12	0.15	0.14	0.10	0.16	0.15	0.13	0.14	0.14	0.12	0.13	0.13
SD	0.04	0.08	0.07	0.04	0.09	0.08	0.10	0.07	0.08	0.05	0.09	0.08
Liberal (L)												
Mean	0.22	0.21	0.21	0.24	0.21	0.22	0.21	0.19	0.19	0.15	0.18	0.17
SD	0.10	0.08	0.08	0.08	0.09	0.09	0.11	0.08	0.09	0.06	0.05	0.06
C+L												
Mean	0.34	0.36	0.35	0.34	0.37	0.37	0.34	0.33	0.33	0.28	0.31	0.30
SD	0.13	0.13	0.13	0.11	0.14	0.13	0.17	0.13	0.14	0.09	0.12	0.11
No-Match Consistent												
Mean	0.04	0.03	0.04	0.03	0.03	0.03	0.03	0.02	0.03	0.02	0.03	0.03
SD	0.02	0.05	0.04	0.03	0.03	0.03	0.04	0.02	0.02	0.01	0.02	0.02
Highly Connected												
Mean	0.44	0.44	0.44	0.43	0.47	0.46	0.43	0.43	0.43	0.34	0.39	0.38
SD	0.15	0.15	0.15	0.11	0.14	0.13	0.19	0.19	0.19	0.07	0.14	0.12
Far Transfer												
Conservative (C)												
Mean	0.13	0.15	0.14	0.15	0.13	0.13	0.13	0.16	0.15	0.14	0.15	0.15
SD	0.08	0.08	0.08	0.08	0.09	0.08	0.10	0.10	0.10	0.12	0.10	0.11
Liberal (L)												
Mean	0.25	0.27	0.26	0.23	0.25	0.25	0.25	0.26	0.26	0.22	0.23	0.23
SD	0.08	0.08	0.08	0.10	0.10	0.10	0.10	0.09	0.09	0.08	0.11	0.10
C+L												
Mean	0.39	0.41	0.41	0.37	0.38	0.38	0.39	0.42	0.41	0.36	0.38	0.38
SD	0.15	0.14	0.14	0.16	0.16	0.16	0.19	0.15	0.16	0.18	0.16	0.17
No-Match Consistent												
Mean	0.03	0.03	0.03	0.01	0.02	0.02	0.03	0.02	0.02	0.04	0.03	0.03
SD	0.02	0.03	0.03	0.02	0.03	0.02	0.02	0.02	0.02	0.04	0.02	0.03
Highly Connected												
Mean	0.60	0.55	0.56	0.53	0.50	0.51	0.59	0.53	0.55	0.47	0.54	0.52
SD	0.20	0.18	0.18	0.22	0.22	0.21	0.21	0.19	0.19	0.20	0.23	0.22

Table 14 ***continued.***

	Causal						General					
	Online			Offline			Online			Offline		
	Paraphrasers (*n* = 7)	Elaborators (*n* = 23)	Total (*n* = 30)	Paraphrasers (*n* = 7)	Elaborators (*n* = 23)	Total (*n* = 30)	Paraphrasers (*n* = 9)	Elaborators (*n* = 19)	Total (*n* = 28)	Paraphrasers (*n* = 9)	Elaborators (*n* = 19)	Total (*n* = 28)
Oral Reading (Words Read Correctly)												
Near Transfer												
Mean	130.86	126.65	127.63	124.00	128.17	127.20	127.11	130.53	129.43	122.78	127.47	125.96
SD	14.51	24.20	22.16	18.00	24.00	22.52	30.55	24.76	26.23	21.98	25.84	24.35
Far Transfer												
Mean	143.57	140.22	141.00	138.57	139.00	138.90	137.67	136.58	136.93	142.67	138.58	137.18
SD	9.40	27.85	24.67	23.61	28.92	27.38	32.15	27.32	28.35	24.67	25.81	25.29

Figure 8. ***Near Transfer Recall by Question Type and Timing, and Reader Subgroup***

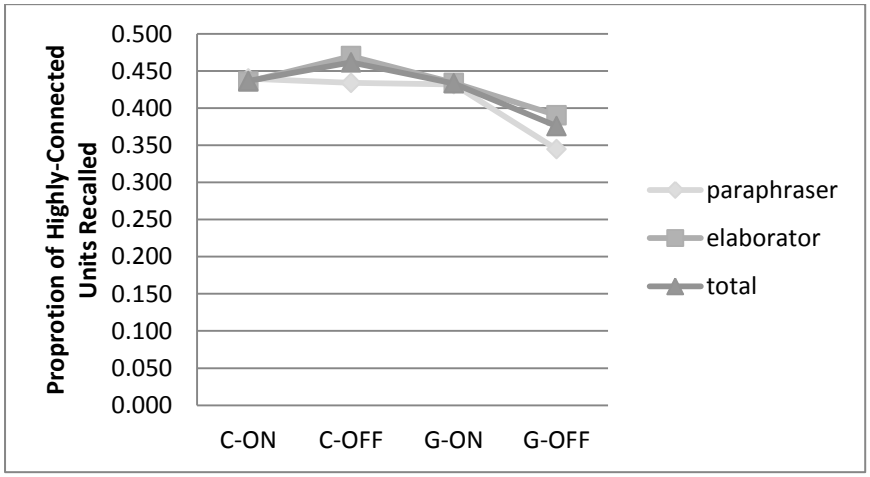
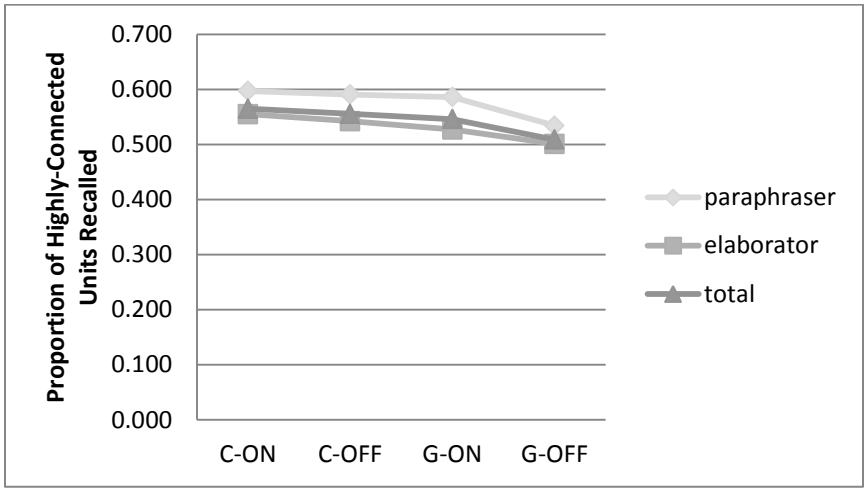


Figure 9. ***Far Transfer Recall by Question Type and Timing, and Reader Subgroup***



Round 5 (Spring Trial Run)

Purpose

The purpose of the Spring Trial Run was to examine the following four variations on the intervention (along with rationale):

(1) **Implementation with informational text**. The rationale is that, in our original proposal, we planned to run parallel studies during the Spring Pilot with both narrative and informational text. However, due to difficulties identifying appropriate informational text (described later in this report), as well as insufficient time to run all we did not run these studies as planned. We decided to move forward with informational text by running a brief (2-week) trial using two texts identified as fitting the criteria described under Developing Texts and Questions.

(2) **One-on-one vs. small-group implementation**. During the Spring Pilot, we noted that a small number of students did not appear to respond well in a small-group setting (e.g., they did not appear engaged, had difficulty answering questions in a group, or exhibited disruptive behaviors). Tutors and/or classroom nominated students whom they thought might benefit from one-on-one instruction.

(3) **Variations on reading aloud versus silently and with or without a model**. During the focus group session, some teachers had expressed concern that students would feel uncomfortable reading aloud, or that this approach would not facilitate comprehension. Although we kept reading aloud as the standard approach in the Spring Pilot, we wondered whether students would be responsive to other approaches (silent reading, reading with a partner, choral reading, reading with a teacher or peer model or preview).

(4) **Reversing the questioning prompt from "Look Back, Think, Connect It" to "Think, Look Back, Connect It."** Tutors observed that many students over-relied on the text to find answers to the questions. However, although the answers were in the text, they were not always explicitly stated, but rather required the student to make an inference. We wondered whether emphasizing the "thinking" part of the process and using "look back" as a way to confirm, rather than find, an answer, might help prevent this overreliance on the text.

Participants

After the Spring Pilot, we invited teachers to volunteer to continue to participate. Based on teachers' willingness, tutor/teacher nominations for one-on-one candidates, and tutor and student schedules, we selected a small number of students to participate in the Spring Trial Run. Students' parents were notified that they would continue to receive intervention for 2 additional weeks. Students remained in their original questioning type condition (Causal or General). Four students (two Causal and two General) were selected for one-on-one intervention. Two groups (one Causal and one General) of four students were selected to continue receiving small-group intervention.

Measure

Once again, tutors completed daily tutoring logs (in Appendix T). They recorded student attendance, the total number of minutes for each session, and the number of questions answers. Then, they rated the extent to which they agreed with statements about the reading and questioning activities, such as "The reading approach used was well-matched to the reading needs of my student(s)," "The text was appropriately interesting and age-appropriate for my student(s)," and "The questions helped my students make relevant connections in the text." Tutors rated each statement from 1 (disagree) to 4 (agree). For any items rated below 4, they Tutors also indicated which reading approach they used in each session (standard oral reading, choral reading, partner reading, whisper reading, repeated reading with tutor as model, repeated reading with peer as model), and why they selected this approach.

Procedures

Tutors met with their students three times per week, for up to 30 min per session, for two weeks. In Week 1, they read the story, "Rainforest is their Home" about Jane Goodall's work with chimpanzees in Gombe National Forest. During the first session, they read using the standard approach (as in the Spring Pilot intervention), except they used the new "Think, Look Back, Connect It" prompt. After the first session, tutors were asked to try alternative approaches to reading (e.g., reading silently) and to note in their daily intervention logs what prompted this decision. At the end of each session (or at least at the end of each week), tutors asked students the following questions: (1) Did you like our reading activity today? (If yes) What did you like about the reading activity? (If no) What did you NOT like about the reading activity? (2) What did you like/not like about the text we read? (3) What did you like/not like about the way we read the text? (Remind them of the reading approach used).

Tutors brought completed logs to weekly meetings to facilitate discussion about how the intervention was going. Each tutor described how the week's sessions had gone, including how well they thought students responded to the informational text and questions, how well they thought individual students were responding to the one-on-one intervention, their impressions of the different reading approaches, and whether they thought the new prompting approach was functioning better than the original approach. The PI and project coordinators took extensive notes during these discussions.

Results

General feedback included the following:

(1) **Implementation with informational text**. Generally, the texts and questions seemed more difficult than did the narrative text (this could partly be due to the fact that the readability levels of the texts were somewhat higher than those for the narrative texts). Answers to the questions were sometimes difficult to find. The vocabulary was more difficult, but pictures seemed to help a lot.
(2) **One-on-one vs. small-group implementation**. Three of the students seemed to respond very well to one-on-one instruction, and one had a very difficult time staying on task (and eventually refused to do the activities).
(3) **Variations on reading aloud versus silently and with or without a model**. Repeated reading went very well across the groups, especially when the adult tutor provided a model.
(4) **Reversing the questioning prompt from "Look Back, Think, Connect It" to "Think, Look Back, Connect It."** Generally, the revised prompt made it easier for the tutors to emphasize the importance of *thinking* of a connection.

This feedback, along with more extensive notes in tutoring logs and meeting minutes, will be used to inform further revisions to the intervention (analysis of these data are still in process).

C. Opportunities for Training and Professional Development the Project Has Provided

Opportunities for training/professional development (PD) to GRAs/URAs and teachers/specialists are describe in Table 15. Professional development opportunities include Knowledge Building activities, in which we provided opportunities for research staff and teachers to learn about important aspects of the grant, Technical Training, in which we provided research staff with training to administer tests and intervention (much of which involves relevant knowledge and skills that are generalizable to their educational careers), and Mentoring (in which project faculty work with individual graduate students to develop their own research questions and analyses using project data).

Table 15. ***Professional Development Opportunities***

Type of PD	Content (Provided By)	Provided To
Knowledge Building	a. Overview/rationale for grant, theoretical and empirical underpinnings (McMaster) b. Intervention design (McMaster) c. Basics of cognitive processing and comprehension (van den Broek) d. Advanced discussion of cognitive theory and intervention development (McMaster & van den Broek)	a. Research Team (in Lab Meetings), Teachers (in Focus Groups) b. Research Team (in Lab Meeting) c. Research Team (in Lab Meeting via Skype) d. Research Team (in Lab Meeting with live visit by co-PI van den Broek)

Technical Training	e. Human subjects and responsible conduct of research training (University of MN) f. Selecting texts and developing questions (White) g. Administering and scoring CBM Maze and Oral Reading (McMaster & White) h. Administering Think Aloud (White) i. Coding Think Aloud (Carlson & Janda) j. Administering and scoring WJ-III ACH (McMaster & Oberg) k. Administering intervention (McMaster) l. Administering and coding recall task (McMaster, Janda, Pinto, van den Broek, & Espin)	e. All research staff f. GRAs g. GRAs h. GRAs i. GRAs and URAs j. GRAs and URAs k. GRAs and URAs l. GRAs and URAs
Mentoring	m. Developing independent research questions and data analysis (McMaster, van den Broek, Espin)	m. GRAs

D. Results Disseminated to Communities of Interest

Results of the project have been shared to the following communities of interest:

(1) University of Minnesota College of Education and Human Development: Poster Presented at annual CEHD Research Day (March 20, 2012)
(2) University of Minnesota Special Education Program Brown Bag Colloquium: Presentation to Special Education faculty and students (April 4th, 2012)
(3) Making the Right Connections Teacher Share (small group meetings with teacher participants throughout the month of May, following completion of the Spring Pilot)

We also plan to share an abbreviated version of this report with our two participating districts, and will offer to meet with interested school personnel to present our Year 1 findings. Conference proposals for posters and presentations are also underway.

E. Accomplishing Goals in the Next Reporting Period

In the next reporting period, we will complete Year 1 analysis, finalize revisions to the current version of the intervention, and prepare presentations and papers for dissemination (on schedule with original plan). In addition, we will address specific Year 2 goals, with no major changes to the goals outlined in the performance agreement.

II. PRODUCTS

A. Publications, conference papers, and presentations
(Presentations only):

McMaster, K., Pinto, V., Janda, B., van den Broek, P., & Espin, C. (March, 2012). Making the right connections to improve the comprehension of struggling readers. Poster presented at the Annual University of Minnesota CEHD Research Day, Minneapolis, MN.

McMaster, K., Janda, B., & Pinto, V. (April, 2012). Under the big top: Balancing theory, evidence, and practice to develop a reading intervention. Presented to the monthly University of Minnesota Special Education Program Brown Bag Colloquium, Minneapolis, MN.

B. Web site(s):
We are in the process of designing a website on which to post descriptions and products related to this project.

C. Technologies or techniques:
none

D. Inventions, patent applications, and/or licenses:
none

E. Other Products
Measures of fit, feasibility, and promise of the intervention (referenced as Appendices A-T in the report) have been developed and are attached. They include the following:

- **a.** Think-Aloud Texts
- **b.** Initial Teacher Questionnaire
- **c.** Text and Question Review Questionnaire
- **d.** Question Development Rubric
- **e.** Sample Intervention Story (Student Copy)
- **f.** Sample Intervention Story (Teacher Copy)
- **g.** Round 1 Feedback Form
- **h.** Round 2 Focus Group Questionnaire
- **i.** Round 3 Fall Trial Run Intervention Log
- **j.** Round 4 Spring Pilot Intervention Log
- **k.** Round 4 Intervention Rating Form
- **l.** Round 4 Intervention Questionnaire
- **m.** Round 4 Sample Fidelity (Causal Online)
- **n.** Round 4 Student Social Validity Questionnaire
- **o.** Round 4 Tutor Social Validity Questionnaire
- **p.** Round 4 Sample Examiner Versions of Causal and General Posttest Protocols
- **q.** Round 4 Sample Examiner Prompting Guides
- **r.** Round 4 Recall Coding Instructions and Sample Glossary
- **s.** Round 4 Recall Coding Sample Parsed Stories
- **t.** Round 5 Spring Trial Run Intervention Log

III. Participants and Other Collaborating Organizations

A. What individuals have worked on the project?

a. Key Personnel (CVs attached):
   i. Principal Investigator: Kristen McMaster
   ii. Co-Principal Investigators: Paul van den Broek & Christine Espin
   iii. Project Coordinators:
      1. Viveca Pinto (Coordinates intervention-related activities)
      2. Bonita Janda (took over discourse-related activities)

b. Consultants (expert consultants whom we have called upon so far for input and feedback related to various aspects of the project):
   i. Discourse Experts
      1. Dr. Art Graesser, University of Memphis
      2. Dr. David Rapp, Northwestern University
      3. Dr. Sarah Carlson, University of Oregon
      4. Dr. Panayiota Kendeou, Neapolis University (Cyprus)
   ii. Intervention Experts
      1. Dr. Douglas Fuchs

c. Graduate Research Assistants (GRAs; all GRAs have worked on 25% or 50% FTE positions for one or two semesters and summer, except Sarah Scholin, who volunteered to be a tutor on the project)

Amy Bethel (Special Education)
Pyung-Gang Jung (Special Education)
Hui-Chen Hsu (Special Education)
Elizabeth Oberg (Special Education)
Joo Ree Richards (School Psychology)
Sarah Scholin (School Psychology)
Martin van Boekel (Learning & Cognition)

d. Undergraduate Research Assistants (URAs; all URAs were junior or senior Psychology majors who worked on the project as volunteers or for course credit)

Amberly Amero
Felicia Behnke
Sara Gustason
Samantha Heindel
Neil Holmberg
Sonya Kuznetsov
Byung Lee
Jaekook Lee
Bridget Regan
Alec Robertson
Aleksandar Zurovac

B. What other organizations have been involved as partners?

a. Bloomington Public Schools
b. Minneapolis Public Schools

C. Have other collaborators or contacts been involved? No

IV. CHANGES/PROBLEMS

A. Changes in approach and reasons for change

The following changes were made to Year 1 activities and emailed to Project Officer Amy Sussman in January, 2012 (original activities from Performance Agreement are listed below, with change described). Dr. Sussman approved these changes in advance:

a. **Identify non-struggling readers (*n* = 6–8)**
We did not end up using non-struggling readers. The original purpose was to include non-struggling readers in trial runs of the intervention. We realized it made more sense to do trial runs with struggling readers.

b. **Conduct pilot studies on the each intervention (narrative and expository text)**
We conducted pilot studies in January-March, focusing on narrative text. We conducted trial runs with expository (informational) text in May. The reason for delaying the pilot studies with informational text is that it was more difficult than anticipated to determine the best way to do the intervention with informational text, because most informational texts that are used in our schools at fourth grade do not have a structure that lends itself to causal questioning. Delaying the informational text pilot also allowed us to more thoroughly explore the question type and timing variables that are of primary interest in this study.

B. Actual or anticipated problems or delays and actions or plans to resolve them:

There was one minor problem early in the project: The original Discourse Project Coordinator, who had expertise in Think-Aloud coding, had to leave the project. To address this problem, we added a consultant, Dr. Sarah Carlson, to help with Think-Aloud coding training, and hired a new Discourse Project Coordinator. Because a steeper-than-expected learning curve was required to conduct the Think-Aloud coding reliably and accurately, we were not able to identify subgroups of struggling readers a priori to intervention. In future years of the project, the aim will be to identify subgroups prior to assignment to intervention condition.

There are no other anticipated problems or delays.

C. Changes that have a significant impact on expenditures:

None.

D. Significant changes in use or care of human subjects, vertebrate animals, and/or biohazards:

None.

E. Change of primary performance site location from that originally proposed:

None

VI. BUDGETARY INFORMATION

Budget Justification

A. Key Personnel

Dr. Kristen McMaster, Principal Investigator, (2.25 Cal. Months AY/.75 summer)
Dr. McMaster has devoted 25% FTE (2.25 cal months AY and .75 cal months summer) to this project. She oversees all major grant activities, including management of the budget; supervision of Site and Intervention coordinators, GRAs, and data collectors; development and revisions of intervention materials and procedures; conducting training of interventionists and teachers; data collection from various sources (experts, teachers, pilot studies); designing pilot studies; data analysis and interpretation. She also prepares reports required by the agency.

B. Other Personnel

Bonita Janda, Site Coordinator (6 cal months over 12 months)
The site coordinator devotes 50% FTE (6 cal months over 12 months) to this project. The person originally in this position, Mary Jane White, left the project in early fall, 2011, and was replaced by a new site coordinator, Bonita Janda. Bonita Janda provides technical assistance for this project under the direct supervision of the PI. Specifically, she supervises GRAs and URAs; participates in development and revisions of intervention materials and procedures; coordinates data collection efforts with experts; coordinates scheduling for screening and other assessments; coordinates scheduling for interventions conducted during trial runs and pilot studies; oversees data transcription, coding, and entry; assists with data analysis and interpretation; and assists with writing reports.

Viveca Pinto, Intervention Coordinator (6 cal. months over 12 months)
Viveca Pinto devotes 50% FTE (6 cal months over 12 months) to this project. She began the project on a graduate research assistant salary base, and switched to a research fellow salary base upon completing her doctorate. Dr. Pinto coordinates development and revisions of intervention materials and procedures; coordinates data collection efforts with teachers, including facilitating focus groups; coordinates and participates in training of interventionists and teachers; conducts interventions and observations during intervention trial runs and pilot studies; assists with data analysis and interpretation; and assists with writing reports.

Graduate Research Assistants assist with development and revisions of intervention materials and procedures; administer and score screening measures and other assessments; serve as interventionists; code Think Aloud protocols and data obtained from questionnaires, surveys, and observations; attend meetings as needed; and assist with data entry and analysis.

Hourly Data Collectors (originally budgeted for 3.72 cal months over 12 months)
We moved the hourly data collector budget line into the Graduate Research Assistant line. Having GRAs dedicated to the project better suited project needs than would hourly data collectors, given the amount of training and time needed to carry out project data collection activities.

C. Travel (budgeted = $4,000 per year + $1,000 mileage reimbursement, Year 1 expense = $7,041) The PI and one coordinator travelled to the IES/SREE conference in Washington, D.C.

at a reimbursed cost of $2,255. Mileage for PI, coordinators and GRAs travel to schools, originally budgeted for $1,000 and included in materials and supplies, was also added to this category. Those expenses totaled $4,786.

D. Other Direct Expenses

Materials and Supplies (budgeted=$10,950 Yr 1, actual=$3,218.
These funds were used for supplies necessary for project implementation (e.g., intervention binders and folders). A laptop was purchased for data analysis; one camcorder to tape interventions sessions (not yet purchased; plans are to purchase in Fall of Year 2); and several digital voice recorders to record focus group and think aloud sessions. Supply expenses have not been as high as anticipated, as it was not necessary to purchase assessments and software that were available through other resources in the department.

Transcription services, originally included in this budget, were not used because we were able to do transcription in house.

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Consulting (budgeted = $6,000/year, actual Yr 1 = $3,500)
Funds were requested to pay each of the following consultants $500/day X up to 2 days each per project period=$1,000.

Dr. Stan Deno, University of Minnesota (emeritus)
Dr. Doug Fuchs, Vanderbilt University
Dr. Art Graesser, University of Memphis
Dr. Joe Jenkins, University of Washington
Dr. Panayiota Kendeous, McGill University
Dr. David Rapp, Northwestern University

We have worked with all consultants except for Drs. Deno and Jenkins; our plan is to solicit their feedback at the end of Year 1 when final revisions are made based on Year 1 results.

Subaward (= Yr 1: $101,661; actual = $33,887) Drs. Paul van de Broek and Christine Espin of Leiden University serve as co-investigators on the project. We are awaiting additional Year 1 invoices for the remaining costs.

Teacher Stipends and Substitute Teacher Costs (budgeted = $24,320 Yr 1, actual = $15,171)
Funds were used to pay teacher stipends and reimburse substitutes needed for focus groups. Substitute costs were not as high as anticipated because 10 teachers did not attend focus groups during school hours (other arrangements were made to ensure they had input).

Student Incentives (budget = $4,000 Yr 1, actual = $1,874)
Funds were used to purchase gel pens and stickers to be used as incentives for intervention. Costs were not as high as expected as we were able to find discounted items for student incentives.

E. Indirect Costs

The University of Minnesota applies a 34% Indirect Cost rate to all direct costs minus graduate research assistant fringe and the amount of the subcontract over $25,000. The on-campus indirect cost rate is being applied given that more than 50% of personnel on this project are housed on-campus.

Changes to the Budget

- There were no changes to the budget that resulted from modifications to project activities .
- There were no changes to the budget that affected our ability to accomplish project activities as described in the performance agreement

- We do not anticipate that funds will be carried over to the next project period. The current balance of $ $155,905 should be spent down by August 31, 2012, because August costs are not reflected in this report, and we are awaiting invoicing from Leiden University.

- There are no anticipated changes in the budget for the next project period.